

# Pricing & Revenue Management

Functional Design & Backend Screens

**TICVAI | Professional Reference Document**

# Table of Contents

Area 10 — Pricing & Revenue Management

Board 1 — Commercial Pricing Foundation

10.1.1 — Commercial Pricing Command Center

10.1.2 — Price List Master Configuration

10.1.3 — Price Category & Rate Type Library

10.1.4 — Rate Structure Builder

10.1.5 — Product & Service Price Assignment

10.1.6 — Package, Bundle & Add-On Pricing

10.1.7 — Market, Venue & Currency Pricing Structure

10.1.8 — Price Hierarchy & Inheritance Configuration

10.1.9 — Price List Templates, Clone & Reuse

10.1.10 — Commercial Pricing Structure Validation

Board 1 — Final Screen Register

Board 2 — Pricing Rules, Eligibility & Effective Dating

10.2.1 — Pricing Rule Command Center

10.2.2 — Customer Segment & Profile Pricing Rules

10.2.3 — Membership & Loyalty Pricing Rules

10.2.4 — Residency, Nationality & Market Pricing Rules

10.2.5 — Channel-Based Pricing Rules

10.2.6 — Location, Venue & Event Pricing Rules

10.2.7 — Quantity, Group & Volume Pricing Rules

10.2.8 — Effective Date, Season & Day-Based Pricing Rules

10.2.9 — Timeslot, Performance & Time-of-Day Pricing Rules

10.2.10 — Pricing Rule Priority, Conflict Resolution & Testing

Board 2 — Final Screen Register

Board 3 — Tax, Fees & Price Calculation Engine

Board 3 — Tax, Fees & Price Calculation Engine

10.3.1 — Tax, Fee & Calculation Command Center

10.3.2 — Tax Profile & Jurisdiction Configuration

10.3.3 — Tax Rule & Treatment Builder

10.3.4 — Fee & Surcharge Library

10.3.5 — Fee Applicability & Charging Rule Builder

10.3.6 — Fee Waiver, Tax Exemption & Exception Rules

10.3.7 — Price Calculation Sequence & Formula Engine

10.3.8 — Currency Precision, Rounding & Monetary Rules

10.3.9 — Price Breakdown, Calculation Simulation & Explainability

10.3.10 — Calculation Validation, Reconciliation & Service Interface

Board 3 — Final Screen Register

Board 1 — Commercial Pricing FoundationWhat prices exist?

Board 2 — Pricing Rules & Eligibility Which price applies?

Board 3 — Tax, Fees & Calculation Engine How do we reach the final payable amount?

Board 4 — Pricing Governance, Publication & Operations

10.4.1 — Pricing Governance Command Center

10.4.2 — Pricing Change Request & Workspace

10.4.3 — Bulk Pricing Update, Import & Mass Maintenance

10.4.4 — Pricing Version & Baseline Management

10.4.5 — Pricing Change Impact Analysis

10.4.6 — Pricing Approval Workflow & Authority Matrix

10.4.7 — Pricing Publication & Effective-Date Scheduler

10.4.8 — Pricing Distribution, Synchronization & Publication Monitor

10.4.9 — Pricing Rollback & Emergency Control Center

10.4.10 — Pricing History, Audit & Compliance Explorer

Board 4 — Final Screen Register

Board 1 — Commercial Pricing Foundation What prices exist?

Board 2 — Pricing Rules & Eligibility Which price applies?

Board 3 — Tax, Fees & Calculation Engine What is the final payable amount?

Board 4 — Pricing Governance, Publication & Operations

Board 5 — Dynamic Pricing, Revenue Optimization & AI How should TICVAI intelligently optimize the price?

Board 5 — Dynamic Pricing Strategy & Rules Engine

10.5.1 — Dynamic Pricing Strategy Command Center

10.5.2 — Dynamic Pricing Strategy Builder

10.5.3 — Demand, Occupancy & Availability Rule Builder

10.5.4 — Booking Velocity & Time-to-Event Rule Builder

10.5.5 — Seasonal, Calendar, Day & Timeslot Dynamic Rules

10.5.6 — Channel, Customer Segment & Location Dynamic Rules

10.5.7 — Dynamic Price Bands, Ladders & Adjustment Matrix

10.5.8 — Dynamic Pricing Guardrails & Commercial Protection

10.5.9 — Dynamic Pricing Automation Policy & Control

10.5.10 — Rule Priority, Conflict Resolution & Dynamic Pricing Test Console

Board 5 — Final Screen Register

Board 6 — AI Signals, Forecasting & Decision Intelligence

10.6.1 — AI Pricing Intelligence Command Center

10.6.2 — Internal Demand & Booking Signal Hub

10.6.3 — Weather Intelligence & Demand Impact Configuration

10.6.4 — Nearby Event, Exhibition & Local Demand Intelligence

10.6.5 — Competitor Pricing & Market Position Intelligence

10.6.6 — Market, Tourism, Holiday & Contextual Signal Hub

10.6.7 — AI Demand Forecasting & Booking Curve Studio

10.6.8 — Price Elasticity & Revenue Response Intelligence

10.6.9 — AI Pricing Recommendation & Explainability Center

10.6.10 — AI Signal Registry, Data Quality & Model Governance

Board 6 — Final Screen Register

Board 5 — Decision Rules

Board 6 — AI Intelligence

Board 7 — Optimization & Execution

Board 7 — Simulation, Automation & Revenue Optimization

Board 5 — Dynamic Pricing Strategy & Rules Engine Defines how prices are allowed to move.

10.7.1 — Revenue Optimization Command Center

10.7.2 — Pricing Simulation Studio

10.7.3 — Scenario Modeling & What-If Analysis

10.7.4 — A/B Pricing Experiment Studio

10.7.5 — Revenue & Demand Impact Forecasting

10.7.6 — AI Recommendation Review & Decision Queue

10.7.7 — Automation Policy & Autonomous Pricing Orchestrator

10.7.8 — Live Dynamic Price Execution & Deployment Monitor

10.7.9 — Dynamic Pricing Performance & Optimization Analytics

10.7.10 — AI Learning, Model Performance & Optimization Feedback

Board 7 — Final Screen Register

Board 1 — Commercial Pricing Foundation Defines the commercial price structures.

Board 2 — Pricing Rules & Eligibility Determines which commercial price applies.

Board 3 — Tax, Fees & Calculation Engine Calculates the final payable amount.

Board 5 — Dynamic Pricing Strategy & Rules Engine Defines how dynamic prices are allowed to move.

# Area 10 — Pricing & Revenue Management

## Board 1 — Commercial Pricing Foundation

The screenshot displays the TICVAI Board 1 - Commercial Pricing Foundation interface, which is organized into 10 screens for configuring pricing and revenue management. The interface includes a sidebar with navigation options like Price Lists, Categories, Rates, Products, Markets, Hierarchy, Templates, Validation, and Settings. The main content area is divided into 10 numbered sections:

- 10.1.1 Commercial Pricing Command Center:** Overview of price lists with metrics like Total Price Lists (86), Active Price Lists (52), Draft Price Lists (18), and Price Categories (68). It also shows Recent Price Lists with details like Rate Name, Rate, Type, Market, Currency, Version, Status, and Modified On.
- 10.1.2 Price List Master Configuration:** Form for creating a new price list, including fields for Price List Name, Price List Code, Price List Type, Legal Entity, Brand, Business Unit, and Description.
- 10.1.3 Price Category & Rate Type Library:** Table listing various price categories and rate types, such as ADULT, CHILD, SENIOR, STUDENT, RESIDENT, MEMBER, VIP, GROUP, CORPORATE, and STAFF, with their respective descriptions and status.
- 10.1.4 Rate Structure Builder:** Form for defining rate structures, including Rate Name, Rate Type, Amount, Unit, Base, Rounding, Status, and A Details.
- 10.1.5 Product & Service Price Assignment:** Form for assigning products and services to a price list, including Product Name, Product Code, Type, Venue, and Assignment Details.
- 10.1.6 Package, Bundle & Add-On Pricing:** Form for creating packages, bundles, and add-ons, including Package Name, Package Code, Package Type, and Package Details.
- 10.1.7 Market, Venue & Currency Pricing Structure:** Form for defining market, venue, and currency pricing structures, including Market Name, Market Code, Market Type, Venue, Currency, and Status.
- 10.1.8 Price Hierarchy & Inheritance Configuration:** Form for configuring price hierarchy and inheritance, including Hierarchy Name, Hierarchy Code, Hierarchy Type, and Hierarchy Details.
- 10.1.9 Price List Templates, Clone & Reuse:** Form for creating and managing price list templates, including Template Name, Template Code, Template Type, and Template Details.
- 10.1.10 Commercial Pricing Structure Validation:** Validation screen showing overall readiness (94%), validation results, and a dependency map.

### 10 Backend Configuration Screens

This is the **restructured Board 1**. It should focus only on the commercial pricing foundation.

It should **not** contain dynamic pricing, tax calculation, fee calculation, approval workflows, or pricing operations. Those belong to later boards.

### Board 1 Objective

Establish the **master commercial pricing architecture** used by every TICVAI sales and distribution channel.

**The fundamental principle is:**

**Products define what is sold. Pricing defines what it costs. Channels determine where it is sold.**

A ticket, membership, F&B item, rental, resource, package, or add-on should therefore **reference a centrally governed price**, rather than storing independent pricing logic inside each module.

**The configuration journey is:**

**Pricing Portfolio → Price List → Price Categories → Rate Structure → Product Assignment → Package Pricing → Market/Currency → Hierarchy → Templates → Validate**

## 10.1.1 — Commercial Pricing Command Center

### Purpose

Provide the central administrative workspace for all commercial pricing structures across TICVAI.

This is the first page a Revenue/Pricing Administrator sees when entering the module.

### KPI Cards

#### Display:

- Total Price Lists
- Active Price Lists
- Draft Price Lists
- Price Categories
- Configured Rates
- Products with Pricing
- Products Missing Pricing
- Markets
- Currencies
- Pricing Validation Issues
- Recently Modified Price Lists
- Upcoming Price Structures

### Price List Directory

#### Each price list should show:

- Price List ID
- Name
- Code
- Type
- Currency
- Market
- Venue
- Brand
- Product Count
- Rate Count
- Effective Period
- Version
- Status
- Owner

### Price List Types

#### Examples:

- Standard Retail
- Venue
- Attraction
- Event

- Membership
- Group
- Corporate
- B2B
- Reseller
- OTA
- Internal
- Special Market

### Status

**Draft / Configured / Validated / Active / Inactive / Expired / Archived**

Approval/publication status will be handled by Board 4.

### Search & Filters

#### Filter by:

- Venue
- Brand
- Market
- Country
- Currency
- Product Type
- Price List Type
- Owner
- Status

### Quick Actions

- Create Price List
- Duplicate
- Open
- Compare
- Validate
- View Dependencies
- Export
- Archive

### AI Assistance

#### AI can identify foundation-level issues such as:

18 active ticket products have no commercial price structure.

Dubai Parks and Abu Dhabi Attractions use different rate-category structures for equivalent products.

### Permissions

#### Separate permissions for:

- View Pricing
- Create Price Lists
- Modify Pricing Structure

- Duplicate
- Archive
- Export

### Acceptance Condition

Authorized administrators can view and manage TICVAI's complete commercial pricing portfolio from one centralized workspace.

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## 10.1.2 — Price List Master Configuration

### Purpose

Create the master container that holds commercial rates.

A **Price List** should be reusable across products and channels.

### General Configuration

#### Configure:

- Price List Name
- Price List Code
- Description
- Price List Type
- Legal Entity
- Brand
- Business Unit
- Country
- Market
- Venue
- Default Currency
- Owner
- Tags
- Status

### Scope

#### Price list can apply to:

- Global
- Country
- Market
- Brand
- Venue
- Event
- Business Unit

### Commercial Types

#### Examples:

**Standard Retail Price List**

**B2B Trade Price List**

**Corporate Price List**

**Membership Price List**

**Group Price List**

**OTA Price List**

**Default Behaviors**

**Configure:**

- Default Rate Category
- Default Currency
- Default Rounding Profile
- Default Price Hierarchy
- Allow Overrides
- Allow Inheritance
- Allow Multiple Currencies
- Allow Product-Specific Rates

**Dependencies**

**Display which modules consume the list:**

**Ticketing / B2C / POS / Kiosk / B2B / Group Sales / Membership / F&B / Retail / Rental**

**Duplicate**

**Administrators can clone:**

UAE Standard 2026

**into:**

UAE Standard 2027

without rebuilding the complete structure.

**Acceptance Condition**

Administrators can create reusable commercial price-list masters with clearly defined ownership, scope, currency, market, and business applicability.

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## **10.1.3 — Price Category & Rate Type Library**

**Purpose**

Define standardized commercial rate categories used across TICVAI.

This avoids different venues independently creating categories such as:

“Adult,” “Adult Standard,” “Normal Adult,” and “Full Adult.”

**Standard Price Categories**

**Support:**

- Adult
- Child
- Junior
- Senior
- Student

- Resident
- Non-Resident
- Member
- VIP
- Group
- Corporate
- B2B
- Complimentary
- Staff
- Promotional
- Custom

### Category Configuration

#### Each category contains:

- Category ID
- Name
- Code
- Description
- Category Family
- Display Name
- Sort Order
- Icon/Label
- Active/Inactive

### Rate Types

Separate **who the price represents** from **how the rate behaves**.

#### Examples:

- Standard
- Reduced
- Contract
- Negotiated
- Complimentary
- Fixed
- Derived
- Package
- Add-On

### Parent/Child Structure

#### Example:

#### Visitor

- Adult
- Child
- Senior

#### Member

- Silver
- Gold
- Platinum

### Reuse

A single category should be reusable across multiple price lists.

### Localization

Support multilingual display names, including Arabic and English.

### AI Standardization

AI can detect duplicate or semantically similar categories.

### Example:

“Kids”, “Child”, and “Children Rate” appear to represent the same commercial category.

### Acceptance Condition

TICVAI maintains a standardized reusable library of commercial price categories and rate types across all tenants, venues, and products.

---

## 10.1.4 — Rate Structure Builder

### Purpose

Define the actual monetary rates contained within a price list.

This is the core commercial configuration screen.

### Rate Matrix

### Example:

| Category    | Rate Type | Amount  |
|-------------|-----------|---------|
| Adult       | Standard  | AED 250 |
| Child       | Reduced   | AED 180 |
| Senior      | Reduced   | AED 190 |
| Resident    | Reduced   | AED 210 |
| Group       | Group     | AED 195 |
| Member Gold | Member    | AED 175 |

### Rate Configuration

### Each rate should contain:

- Rate ID
- Rate Name

- Rate Code
- Price Category
- Rate Type
- Amount
- Currency
- Unit Basis
- Precision
- Rounding Profile
- Status

### Unit Basis

#### Support:

- Per Ticket
- Per Person
- Per Unit
- Per Hour
- Per Day
- Per Session
- Per Resource
- Per Package
- Per Membership Period

This is important because TICVAI pricing extends beyond ticketing.

### Derived Rates

Allow rates to be based on another commercial rate.

#### Example:

**Child = Adult – 25%**

**or:**

**VIP = Standard + AED 200**

However, this remains a **commercial rate relationship**, not dynamic pricing.

### Matrix Editing

#### Administrators should be able to:

- Add Rate
- Duplicate Rate
- Edit
- Disable
- Sort
- Copy Across Categories
- Compare

### Validation

#### Detect:

- Duplicate Rates
- Missing Amounts

- Unsupported Currency
- Invalid Derived Rate
- Circular Rate Relationship

### Acceptance Condition

Administrators can define monetary rate structures for all TICVAI commercial product types through configurable rate matrices.

---

## 10.1.5 — Product & Service Price Assignment

### Purpose

Connect commercial rates to the actual products and services being sold.

### Supported Commercial Objects

#### Pricing can be assigned to:

- Ticket Product
- Ticket Type
- Admission
- Event
- Performance
- Membership
- Annual Pass
- Add-On
- F&B Item
- Retail Product
- Rental Item
- Resource
- Reservation Service
- Experience
- Other Sellable Service

### Assignment Workspace

#### Show:

**Product → Price List → Category → Rate**

#### Example

#### Theme Park — 1 Day Admission

Adult → AED 250

Child → AED 180

Resident → AED 210

Group → AED 195

#### Bulk Assignment

#### Select:

25 attraction products

#### and assign:

UAE Standard Admission Price List.

## Assignment Scope

### Allow:

- Product Level
- Product Variant
- Ticket Type
- Event
- Performance
- Venue

## Price Source Visibility

### Product configuration should display:

#### Pricing Source: UAE Standard Admission 2027

rather than duplicating the monetary value inside the product.

## Dependency View

Show all consuming products for a selected rate.

### Example:

Adult Standard AED 250 is referenced by 17 products.

## Acceptance Condition

All sellable TICVAI objects can reference centrally managed price lists and rates without maintaining duplicate price definitions inside business modules.

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## 10.1.6 — Package, Bundle & Add-On Pricing

### Purpose

Provide dedicated commercial structures for products containing multiple components.

This is separate from the Product Relationship/Bundle module.

**Product Catalogue defines what the bundle contains. Pricing defines how that bundle is priced.**

## Package Pricing Models

### Support:

- Fixed Package Price
- Sum of Components
- Discounted Component Sum
- Component Override
- Included Component
- Optional Paid Component

### Example

#### Family Experience Package

2 Adult Tickets = AED 500

2 Child Tickets = AED 360

Meal Voucher = AED 120

### Normal Total:

**AED 980**

**Package Price:****AED 850****Commercial package saving:****AED 130****Add-On Pricing****Configure pricing for:**

- Fast Track
- Parking
- Meal
- Photo
- Equipment
- Upgrade
- Additional Session
- Premium Access

**Required vs Optional****Pricing engine must understand:****Included****Required Paid****Optional Paid****Component Price Visibility****Configure whether the customer sees:**

- Package Total Only
- Individual Components
- Component + Package Saving

**Bundle Price Integrity**

System should validate that component changes do not unintentionally break a fixed or derived package price.

**Acceptance Condition**

Packages, bundles, and add-ons can use centralized commercial pricing while their product composition remains owned by the Product Catalogue.

## 10.1.7 — Market, Venue & Currency Pricing Structure

**Purpose**

Support TICVAI's multi-country, multi-market, multi-venue and multi-currency commercial model.

**Market Hierarchy****Example:****Global**

- UAE
- Dubai
- Venue A

**or:**

### **Global**

- Saudi Arabia
- Riyadh
- Venue B

### **Pricing Dimensions**

#### **Configure:**

- Country
- Market
- Region
- Venue
- Brand
- Legal Entity
- Currency

### **Currency Configuration**

#### **Support:**

- Base Currency
- Selling Currency
- Currency Precision
- Rounding
- Display Format

### **Example**

#### **Same product:**

UAE Price List → AED 250

Saudi Price List → SAR 255

UK Price List → GBP 55

These should be **independently governed commercial prices**, not necessarily simple FX conversions.

### **FX-Assisted Setup**

#### **TICVAI may show an FX reference during setup:**

AED 250 ≈ SAR 255

but the commercial administrator decides the final selling price.

### **Venue Overrides**

#### **Example:**

Global Adult Rate

→ AED 250

Dubai Venue Override

→ AED 275

Abu Dhabi Venue

→ Inherit AED 250

### **Acceptance Condition**

TICVAI can maintain controlled commercial pricing across multiple markets, venues, legal entities, and currencies.

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## 10.1.8 — Price Hierarchy & Inheritance Configuration

### Purpose

Define where TICVAI should obtain a price when multiple commercial pricing layers exist. This is essential to prevent conflicting prices.

### Example Hierarchy

#### Global Master Price

↓

#### Country Price

↓

#### Market Price

↓

#### Venue Price

↓

#### Product Price

↓

### Approved Specific Override

#### Hierarchy Builder

#### Administrators configure:

- Hierarchy Level
- Priority
- Inheritance
- Override Permission
- Fallback Behavior

### Example

Global Adult = AED 250

Dubai Market = AED 260

Venue A Override = AED 275

Venue B = inherits AED 260.

### Override Rules

#### Configure:

- Override Allowed
- Override Requires Reason
- Maximum Override Range
- Override Expiry
- Return to Parent Price

Approval of overrides belongs to Board 4.

### Fallback

**If a child rate does not exist:****Use Parent / Use Default / Block Sale****Conflict Detection****TICVAI should identify:**

Two price sources have equal priority for the same product.

**Price Resolution Preview****Administrator can ask:**

What price source would Product X use at Venue Y?

**TICVAI responds:**

AED 275 — Venue A Commercial Price List.

**Acceptance Condition**

The platform deterministically resolves the correct commercial price source across global, market, venue, and product hierarchies.

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## 10.1.9 — Price List Templates, Clone & Reuse

**Purpose**

Accelerate commercial setup across new venues, events, seasons and markets.

**Template Library****Examples:**

- Standard Attraction Pricing
- Theme Park Pricing
- Museum Pricing
- Concert Pricing
- Sports Pricing
- Membership Pricing
- Group Pricing
- Corporate Pricing
- B2B Pricing
- Rental Pricing

**Template Components****Templates may include:**

- Price Categories
- Rate Types
- Rate Matrix Structure
- Currency Structure
- Hierarchy
- Product Mapping Pattern
- Package Pricing Pattern

**Clone****Allow cloning:**

- Complete Price List
- Rate Structure Only
- Selected Categories
- Product Assignments
- Market Structure

### New Season Wizard

#### Example:

#### Create 2027 Pricing from 2026

#### Options:

- ☒ Copy categories
- ☒ Copy rate structure
- ☒ Copy product mapping
- ☐ Copy monetary values
- ☒ Increase values by 5%

The actual bulk price operation will be governed in Board 4.

### AI-Assisted Template Generation

#### AI can recommend a starting structure based on:

- Venue Type
- Product Portfolio
- Similar Venues
- Existing TICVAI Configuration

#### Example:

“Create a standard theme-park pricing structure.”

#### AI drafts:

Adult / Child / Resident / Group / Member / VIP

for administrator review.

### Acceptance Condition

Administrators can rapidly create consistent commercial pricing structures by cloning and reusing approved templates instead of rebuilding pricing manually.

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## 10.1.10 — Commercial Pricing Structure Validation

### Purpose

Validate that the commercial pricing foundation is structurally complete before it proceeds to rule configuration, governance, or publication.

This is **not the final publication screen**. Board 4 owns approval and publication.

### Validation Categories

#### Price List

- Required fields complete
- Currency defined
- Ownership assigned

#### Rate Structure

- Categories configured

- Monetary values valid
- Derived relationships valid

### Product Mapping

- Required products priced
- No orphan assignments

### Package

- Component pricing valid

### Market

- Currency and venue configuration valid

### Hierarchy

- No conflicting source priority
- Fallback configured

### Validation Results

#### Classify:

#### Critical

Sale cannot proceed.

#### Warning

Configuration should be reviewed.

#### Information

Optimization suggestion.

#### Example

- 7 active ticket products have no Adult rate.
- Venue B has no currency configured.
- 14 products inherit pricing from Global because no market rate exists.
- AI identified three duplicate commercial price categories.

### Dependency Map

#### Visualize:

#### Price List

→ Rate

→ Product

→ Market/Venue

→ Consuming Channels

### Readiness Score

#### Example:

#### Commercial Pricing Readiness: 94%

Critical Issues: 2

Warnings: 7

### AI QA

#### AI can identify unusual structures such as:

Child price is 18% higher than Adult price for Product X.

It should flag this rather than automatically changing it.

**Handoff****Once structurally valid:****Ready for Pricing Rules & Eligibility — Board 2****Acceptance Condition**

TICVAI prevents structurally incomplete or contradictory commercial pricing configurations from progressing downstream without clearly identifying the issue.

**Board 1 — Final Screen Register**

| #              | Screen                                      | Responsibility          |
|----------------|---------------------------------------------|-------------------------|
| <b>10.1.1</b>  | Commercial Pricing Command Center           | Pricing portfolio       |
| <b>10.1.2</b>  | Price List Master Configuration             | Price-list master       |
| <b>10.1.3</b>  | Price Category & Rate Type Library          | Commercial categories   |
| <b>10.1.4</b>  | Rate Structure Builder                      | Monetary rate structure |
| <b>10.1.5</b>  | Product & Service Price Assignment          | Product-price linkage   |
| <b>10.1.6</b>  | Package, Bundle & Add-On Pricing            | Composite pricing       |
| <b>10.1.7</b>  | Market, Venue & Currency Pricing Structure  | Multi-market pricing    |
| <b>10.1.8</b>  | Price Hierarchy & Inheritance Configuration | Price-source resolution |
| <b>10.1.9</b>  | Price List Templates, Clone & Reuse         | Reusable pricing setup  |
| <b>10.1.10</b> | Commercial Pricing Structure Validation     | Structural QA           |

**Board Boundary****To keep the architecture clean, Board 1 deliberately does NOT own:**

**Board 2:** customer eligibility, membership/segment/channel/location rules, effective dates, seasons, quantity rules and rule priority.

**Board 3:** taxes, fees, surcharges, exemptions, calculation sequencing and final payable calculation.

**Board 4:** bulk changes, approvals, versioning, publication, rollback, history and audit.

**Board 5:** dynamic pricing, demand forecasting, elasticity, simulation, A/B testing and AI revenue optimization.

**That gives us a much stronger foundation:**

**Board 1 answers: “What are our commercial prices and how are they structured?”**

**The next Board will answer:**

**“Who receives which price, where, when, and under what conditions?”**

## Area 10 — Pricing & Revenue Management

# Board 2 — Pricing Rules, Eligibility & Effective Dating

The screenshot displays the TICVAI Board 2 interface, titled "AREA 10 - PRICING & REVENUE MANAGEMENT | BOARD 2 - PRICING RULES, ELIGIBILITY & EFFECTIVE DATING (10 SCREENS)". The interface is divided into a sidebar and a main content area with 10 numbered screens.

- Screen 1: 10.2.1 Pricing Rule Command Center** - Overview of rule counts (Active: 326, Draft: 42, Rule Conflicts: 7) and a table of recent rules.
- Screen 2: 10.2.2 Customer Segment & Profile Pricing Rules** - Configuration for rules like "VIP Customer Rate".
- Screen 3: 10.2.3 Membership & Loyalty Pricing Rules** - Configuration for rules like "Gold Member Pricing".
- Screen 4: 10.2.4 Residency, Nationality & Market Pricing Rules** - Configuration for rules like "UAE Resident Pricing".
- Screen 5: 10.2.5 Channel-Based Pricing Rules** - Configuration for rules like "B2B Trade Channel Pricing".
- Screen 6: 10.2.6 Location, Venue & Event Pricing Rules** - Configuration for rules like "Dubai Parks Venue Pricing".
- Screen 7: 10.2.7 Quantity, Group & Volume Pricing Rules** - Configuration for rules like "Group 10+ Discount".
- Screen 8: 10.2.8 Effective Date, Season & Day-Based Rules** - Configuration for rules like "Peak Season Pricing".
- Screen 9: 10.2.9 Timeslot, Performance & Time-of-Day Pricing Rules** - Configuration for rules like "Evening Timeslot Pricing".
- Screen 10: 10.2.10 Pricing Rule Priority, Conflict Resolution & Testing** - A detailed simulation and conflict resolution tool showing rule evaluation results and conflict alerts.

## 10 Backend Configuration Screens

### Board Objective

Board 1 defined **what the commercial prices are and how they are structured.**

### Board 2 defines:

**Who gets which price, where, when, and under what conditions?**

This board is the decision layer between the commercial price foundation and the final calculation engine.

### It should govern contextual pricing based on:

**Customer → Membership → Loyalty → Residency → Market → Channel → Location → Quantity → Date/Season → Day/Timeslot → Priority → Resolution**

### The core rule is:

**Board 2 selects the applicable commercial rate. It does not calculate tax/fees and does not perform dynamic pricing.**

## 10.2.1 — Pricing Rule Command Center

### Purpose

Provide administrators with a centralized workspace for all pricing eligibility and contextual pricing rules.

## KPI Cards

### Display:

- Active Pricing Rules
- Draft Rules
- Customer Segment Rules
- Membership Rules
- Residency Rules
- Channel Rules
- Location Rules
- Quantity Rules
- Seasonal Rules
- Timeslot Rules
- Rule Conflicts
- Rules Expiring Soon

## Rule Directory

### Each rule should show:

- Rule ID
- Rule Name
- Rule Type
- Price List
- Product Scope
- Customer Scope
- Channel
- Venue/Location
- Effective From
- Effective To
- Priority
- Status
- Owner

## Rule Types

### Support:

- Customer Segment
- Membership
- Loyalty
- Residency
- Nationality
- Channel
- Location
- Quantity
- Group
- Seasonal

- Day-of-Week
- Timeslot
- Event
- Corporate/B2B
- Custom

### Quick Actions

- Create Rule
- Duplicate
- Test Rule
- Compare
- Disable
- View Conflicts
- View Dependencies

### AI Capability

#### AI can identify:

Three rules may return different prices for the same Adult ticket on Saturday through B2C.

#### Acceptance Condition

Administrators can identify, search, and manage all contextual pricing rules from one workspace.

---

## 10.2.2 — Customer Segment & Profile Pricing Rules

### Purpose

Define price eligibility based on customer characteristics and commercial segments.

### Supported Dimensions

#### Configure by:

- Customer Type
- Customer Segment
- Account Type
- CRM Segment
- VIP Status
- Corporate Customer
- Employee/Staff
- Partner Customer
- Guest/Registered User

### Example

#### VIP Customer Rate

#### IF:

Customer Segment = VIP

#### THEN:

Use Rate = VIP Adult

## Customer Segment Sources

### Rules may consume segments from:

- CRM
- Membership
- B2B Partner
- Corporate Account
- Customer Profile

## Rule Configuration

### Capture:

- Rule Name
- Segment
- Applicable Products
- Price List
- Rate
- Priority
- Effective Dates
- Status

### Fallback

#### If customer no longer qualifies:

#### Use Standard Rate

#### AI Capability

AI can detect segment overlap and inconsistent rate treatment.

#### Acceptance Condition

TICVAI can automatically select approved pricing based on governed customer segmentation.

---

## 10.2.3 — Membership & Loyalty Pricing Rules

### Purpose

Control member-specific and loyalty-tier pricing.

### Membership Dimensions

#### Support:

- Membership Product
- Membership Status
- Membership Tier
- Annual Pass Type
- Membership Level
- Active/Suspended Status

### Loyalty Dimensions

#### Support:

- Loyalty Tier

- Loyalty Program
- Points Band
- Customer Status

**Example****Gold Member****IF:**

Membership = Active

AND

Tier = Gold

**THEN:**

Use Gold Member Rate.

**Tiered Pricing****Example:**

Standard → AED 250

Silver → AED 225

Gold → AED 200

Platinum → AED 180

**Member + Guest Rules****Configure whether benefits apply to:**

- Member Only
- Member + 1 Guest
- Member + Family
- Selected Quantity

**Validation****System should detect:**

- Inactive Membership
- Expired Membership
- Missing Tier Rate
- Conflicting Loyalty/Member Rates

**Architecture**

Membership owns eligibility status.

Pricing owns the monetary rate selection.

**Acceptance Condition**

TICVAI can consistently apply member and loyalty pricing without duplicating membership entitlement logic.

## 10.2.4 — Residency, Nationality & Market Pricing Rules

**Purpose**

Support geographically differentiated commercial pricing.

**Eligibility Inputs****Configure by:**

- Residency
- Nationality
- Country
- Market
- Region
- Customer Address
- Verified ID
- Government ID Integration where applicable

**Example****UAE Resident Rate****IF:**

Residency = UAE

AND

Verification = Valid

**THEN:**

Use UAE Resident Adult Rate.

**Market Pricing****Examples:**

- UAE Resident
- GCC Resident
- International Visitor
- Local Community
- Tourism Market
- Domestic Market

**Verification Requirements****Configure whether eligibility requires:**

- Customer Declaration
- Account Profile
- ID Upload
- Government/Identity Verification
- Staff Verification

**Fallback****If eligibility cannot be verified:**

Use Non-Resident Rate.

**Audit**

Record what evidence/rule caused the rate to be selected.

**Acceptance Condition**

Residency, nationality, and market-specific pricing can be applied only when the configured eligibility criteria are satisfied.

## 10.2.5 — Channel-Based Pricing Rules

### Purpose

Determine which commercial rate applies based on sales channel.

### Supported Channels

#### Examples:

- B2C Web
- Mobile App
- POS
- Mobile POS
- Kiosk
- Call Center
- B2B
- Reseller
- OTA
- API
- Partner Portal

### Example

#### 1-Day Admission

B2C → Retail Adult AED 250

POS → Retail Adult AED 250

B2B → Trade Adult AED 205

OTA → Contract Adult AED 215

### Rule Configuration

#### Capture:

- Channel
- Product
- Price List
- Rate
- Market
- Venue
- Effective Dates
- Priority

### Relationship with Sales Channel Management

#### Sales Channel Management determines:

Which pricing profile is associated with the channel.

#### Board 2 determines:

Which contextual rate within the approved pricing structure applies.

### Channel Override

#### Controlled channel pricing overrides can be configured with:

- Start Date

- End Date
- Reason
- Fallback Rate

Approval belongs to Board 4.

### **Acceptance Condition**

Each authorized sales channel consistently receives the correct governed commercial rate.

---

## **10.2.6 — Location, Venue & Event Pricing Rules**

### **Purpose**

Allow rates to vary according to where and for which event/experience the product is sold.

### **Pricing Context**

#### **Support:**

- Country
- City
- Venue
- Attraction
- Location
- Event
- Performance
- Zone
- Experience

### **Example**

#### **Adult Admission:**

Global Rate → AED 250

Dubai Venue → AED 275

Abu Dhabi Venue → AED 250

Special Event → AED 320

#### **Event Override**

Specific event pricing may supersede venue pricing.

#### **Recommended resolution:**

#### **Specific Event**

→ **Venue**

→ **Market**

→ **Global**

#### **Temporary Location Pricing**

#### **Support:**

- Pop-up Venue
- Exhibition
- Seasonal Site
- Temporary Event Location

## Location Conditions

### Configure:

- Venue
- Event
- Product
- Rate
- Effective Dates
- Priority

### Acceptance Condition

TICVAI can resolve the correct commercial rate based on the applicable venue, location, event, or performance context.

---

## 10.2.7 — Quantity, Group & Volume Pricing Rules

### Purpose

Support commercial rates based on purchased quantity or group size.

### Quantity Models

#### Support:

- Minimum Quantity
- Quantity Bands
- Group Size
- Volume Threshold
- Buy-X Rate
- Per-Person Group Rate

### Example

#### Adult Admission:

1-9 → AED 250

10-24 → AED 225

25-49 → AED 210

50+ → AED 195

### Group Rules

#### Support:

- Group Minimum
- Maximum Group Size
- Group Type
- School
- Corporate
- Tour
- Family
- B2B
- Custom

## Tiered Rate Matrix

| Quantity | Rate    |
|----------|---------|
| 1-9      | AED 250 |
| 10-24    | AED 225 |
| 25-49    | AED 210 |
| 50+      | AED 195 |

## Mixed Products

### Allow rules to define whether quantity is calculated:

- Per Product
- Per Product Family
- Per Order
- Per Group Booking

## Complimentary Logic

### Example:

1 complimentary coordinator ticket per 20 paid students.

The Product/Entitlement layer creates the entitlement; Pricing determines the commercial value/qualification rule.

## Acceptance Condition

TICVAI automatically selects appropriate group and volume pricing based on configured quantity thresholds and customer context.

## 10.2.8 — Effective Date, Season & Day-Based Pricing Rules

### Purpose

Control commercial price selection over time.

### Effective Dating

#### Configure:

- Effective From
- Effective To
- Sales Start
- Sales End
- Visit Date Range
- Season
- Holiday Period
- Event Period

**Seasonal Pricing****Examples:****Low Season**

AED 200

**Regular Season**

AED 250

**Peak Season**

AED 290

**Day-of-Week Pricing****Example:**

Monday–Thursday → AED 220

Friday → AED 250

Saturday–Sunday → AED 280

**Calendar Rules****Support:**

- Public Holidays
- School Holidays
- Special Dates
- Blackout Dates
- Peak Dates

**Calendar View**

Administrators should visualize pricing by date.

**Overlap Detection****System should detect:**

Peak Season and Public Holiday rules overlap with different rates.

**Acceptance Condition**

TICVAI consistently selects date-appropriate pricing based on effective periods, seasons, and day-based rules.

**10.2.9 — Timeslot, Performance & Time-of-Day Pricing Rules****Purpose**

Allow commercial pricing to differ across times within the same day or event.

**Supported Contexts**

- Timeslot
- Performance
- Start Time
- Arrival Window
- Session
- Time-of-Day Band

**Example**

**Museum Admission:**

09:00–12:00 → AED 180

12:00–16:00 → AED 220

16:00–20:00 → AED 200

**Peak/Off-Peak****Configure:**

- Peak
- Standard
- Off-Peak
- Late Entry
- Early Entry

**Performance Pricing****Example:****Concert:**

Matinee → AED 350

Evening → AED 450

Final Performance → AED 550

**Rule Configuration****Capture:**

- Product
- Event
- Performance
- Timeslot
- Price List
- Rate
- Time Range
- Effective Dates
- Priority

**Validation****Detect:**

- Overlapping time ranges
- Missing timeslot rate
- Conflicting performance rule

**Acceptance Condition**

TICVAI can apply different governed commercial rates according to timeslot, performance, or time-of-day context.

## 10.2.10 — Pricing Rule Priority, Conflict Resolution & Testing

**Purpose**

Define how TICVAI decides the final applicable rate when multiple pricing rules match.

This is the critical control screen for Board 2.

## **Rule Resolution**

### **Recommended hierarchy:**

#### **Contract/Partner Rule**

↓

#### **Specific Customer/Member Rule**

↓

#### **Event/Performance Rule**

↓

#### **Venue/Location Rule**

↓

#### **Channel Rule**

↓

#### **Quantity/Group Rule**

↓

#### **Season/Date Rule**

↓

#### **Standard Rate**

This hierarchy should be configurable.

## **Priority Configuration**

### **For every rule:**

- Priority
- Specificity
- Stop Processing
- Continue Processing
- Override Allowed
- Fallback Behavior

## **Conflict Detection**

### **Identify:**

- Same-priority matches
- Contradictory rates
- Circular fallback
- Unreachable rule
- Missing fallback
- Overlapping time/date conditions

## **Pricing Rule Simulator**

### **Inputs:**

- Customer
- Membership
- Residency
- Product

- Quantity
- Venue
- Channel
- Date
- Timeslot

### Example

#### Input:

Customer: Gold Member  
 Residency: UAE  
 Product: Adult Admission  
 Channel: B2C  
 Venue: Dubai  
 Date: Saturday  
 Time: 18:00

### Simulation Result

#### Candidate Rates:

- Standard Adult — AED 250
- UAE Resident — AED 220
- Saturday Peak — AED 280
- Gold Member — AED 190

#### Resolved Rate:

**Gold Member — AED 190**

#### Reason:

Member pricing priority exceeds resident, day, and standard pricing.

### Explainability

#### Show:

**Rules Evaluated → Rules Matched → Rules Rejected → Priority → Final Rate**

### AI Capability

AI can recommend simplification where unnecessary rule complexity exists.

#### Example:

11 rules produce the same final rate across all tested scenarios and may be consolidated.

### Acceptance Condition

For any commercial transaction, TICVAI can deterministically resolve and explain exactly why a specific rate was selected.

## Board 2 — Final Screen Register

| #      | Backend Screen              | Primary Responsibility    |
|--------|-----------------------------|---------------------------|
| 10.2.1 | Pricing Rule Command Center | Contextual rule portfolio |

| #       | Backend Screen                                       | Primary Responsibility |
|---------|------------------------------------------------------|------------------------|
| 10.2.2  | Customer Segment & Profile Pricing Rules             | Customer-based pricing |
| 10.2.3  | Membership & Loyalty Pricing Rules                   | Member/loyalty rates   |
| 10.2.4  | Residency, Nationality & Market Pricing Rules        | Geographic eligibility |
| 10.2.5  | Channel-Based Pricing Rules                          | Channel pricing        |
| 10.2.6  | Location, Venue & Event Pricing Rules                | Location/event pricing |
| 10.2.7  | Quantity, Group & Volume Pricing Rules               | Volume pricing         |
| 10.2.8  | Effective Date, Season & Day-Based Pricing Rules     | Date/season pricing    |
| 10.2.9  | Timeslot, Performance & Time-of-Day Pricing Rules    | Time-based pricing     |
| 10.2.10 | Pricing Rule Priority, Conflict Resolution & Testing | Final rate resolution  |

## Board 2 Architecture

### The decision journey is:

**Customer → Membership → Residency → Channel → Location/Event → Quantity → Date/Season → Timeslot → Priority → Final Rate**

### Critical separation

### Board 2 stops once TICVAI has resolved:

**“The applicable commercial rate is AED 190.”**

### It does not yet answer:

“What tax applies?”

“What service fee applies?”

“What is the final payable amount?”

### Those belong to:

## Board 3 — Tax, Fees & Price Calculation Engine

**So the pricing architecture remains clean:**

## Area 10 — Pricing &amp; Revenue Management

## Board 3 — Tax, Fees &amp; Price Calculation Engine

The screenshot displays the TICVAI Board 3 interface, titled "AREA 10 - PRICING & REVENUE MANAGEMENT | BOARD 3 - TAX, FEES & PRICE CALCULATION ENGINE (10 SCREENS)". The interface is divided into 10 numbered screens:

- 10.3.1 Tax, Fee & Calculation Command Center:** Overview of Tax Profiles (28), Jurisdictions (14), Fee Profiles (42), and Surcharges (17). It includes a table for "Recent Configuration Activity" with columns for Type, Name, Entry / Venue, Status, Modified On, and Modified By.
- 10.3.2 Tax Profile & Jurisdiction Configuration:** Search for tax profiles by Country, Tax Type, Rate Hs, Effective From, Effective To, Status, and Actions.
- 10.3.3 Tax Rule & Treatment Builder:** Rule Details (Rule Name, Condition, Product Category, Venue, Channel, Customer Type, Tax Treatment, Exclusion) and Conditions (Condition, Operator, Value).
- 10.3.4 Fee & Surcharge Library:** Search for fees by Fee Name, Fee Code, Type, Method, Value, Tax Treatment, and Status.
- 10.3.5 Fee Applicability & Charging Rule Builder:** Rule Details (Rule Name, Condition, Operator, Value) and Conditions (Condition, Operator, Value).
- 10.3.6 Fee Waiver, Tax Exemption & Exception Rules:** Exception Details (Rule Name, Condition, Operator, Value) and Conditions (Condition, Operator, Value).
- 10.3.7 Price Calculation Sequence & Formula Engine:** Calculation Sequence Steps (1. Base Price (Discount Field), 2. Discounts / Promotions, 3. Package Adjustment, 4. Fees / Surcharges, 5. Taxes, 6. Rounding, 7. Final Amount) and Formula Preview.
- 10.3.8 Currency Precision, Rounding & Monetary Rules:** Currency Precision Rules (Currency, Code, Rounding Method, Rounding Method, Effective From, Effective To) and Rule Details.
- 10.3.9 Price Breakdown, Calculation Simulation & Explainability:** Simulation Inputs (Customer, Quantity, Step Code, Step Description) and Calculation Breakdown (Step, Selected Rate, Cost/Membership, Amount, Amount (PES), Amount (PES), Amount (PES)).
- 10.3.10 Calculation Validation, Reconciliation & Service Interface:** Validation Summary (Overall Health: 98%, Passed: 142, Warnings: 6, Errors: 2, Critical: 0) and Calculation Service Interface (Status: Success, Version: v2.0.0, Auth Type: OAuth 2.0, Rate Limit: 1000 req/min).

## 10 Backend Configuration Screens

## Board Objective

Board 1 established **what commercial prices exist**.

Board 2 determines **which commercial rate applies** to a transaction.

## Board 3 now determines:

**How does TICVAI transform the selected commercial rate into the final payable amount?**

This board should become the **central calculation engine** for the entire TICVAI platform.

## It owns:

**Tax → Fees → Surcharges → Waivers/Exemptions → Calculation Sequence → Rounding → Price Breakdown → Simulation → Validation → Calculation API**

**It should serve every selling channel consistently:**

**B2C / POS / Mobile POS / Kiosk / B2B / OTA / Call Center / Group Sales / Membership / F&B / Retail / Rental**

## Critical Architecture

No sales channel should independently calculate tax or fees.

For the same transaction context, TICVAI must produce the same governed calculation regardless of where the sale originates.

## 10.3.1 — Tax, Fee & Calculation Command Center

### Purpose

Provide Finance, Commercial and Pricing administrators with one central view of TICVAI's price-calculation configuration.

### KPI Cards

#### Display:

- Active Tax Profiles
- Tax Jurisdictions
- Active Fee Profiles
- Active Surcharges
- Exemption Rules
- Calculation Profiles
- Products Missing Tax
- Products Missing Calculation Profile
- Configuration Conflicts
- Upcoming Tax Changes
- Validation Issues
- Recently Modified Rules

### Configuration Directory

#### Display:

- Profile Name
- Type
- Country
- Legal Entity
- Market
- Currency
- Product Scope
- Effective Period
- Status
- Owner

### Configuration Types

#### Filter by:

- Tax
- Fee
- Surcharge
- Waiver
- Exemption
- Calculation Profile
- Rounding Profile

### Alerts

#### Examples:

14 active products have no applicable VAT profile.

UAE VAT profile changes effective 1 January.

Two service-fee rules apply to the same B2C transaction.

### **AI Assistance**

AI should identify unusual or potentially conflicting configurations, but never autonomously change tax configuration.

### **Quick Actions**

- Create Tax Profile
- Create Fee
- Create Calculation Profile
- Run Simulation
- Validate Configuration
- View Dependencies

### **Acceptance Condition**

Authorized administrators can understand the complete status and health of TICVAI tax, fee and price-calculation configuration from one workspace.

---

## **10.3.2 — Tax Profile & Jurisdiction Configuration**

### **Purpose**

Define reusable tax profiles according to legal entity, country, jurisdiction and commercial context.

### **Tax Profile Fields**

#### **Configure:**

- Tax Profile Name
- Tax Profile Code
- Tax Type
- Country
- Region/Jurisdiction
- Legal Entity
- Tax Registration Number
- Currency
- Effective From
- Effective To
- Status
- Owner

### **Tax Types**

#### **Support:**

- VAT
- GST
- Sales Tax
- Entertainment Tax

- Tourism Tax
- Municipality Tax
- Service Tax
- Custom Regulatory Tax

### Applicability

#### Apply tax profiles by:

- Legal Entity
- Country
- Market
- Venue
- Product Category
- Product
- Service
- Channel where legally applicable

### Example

#### UAE VAT

Country: UAE

Tax Type: VAT

Rate: 5%

Legal Entity: TICVAI Venue Operator LLC

#### Jurisdiction Hierarchy

#### Support:

#### Country

→ **Region**

→ **Municipality**

where different rules apply.

#### Effective Dating

New tax structures must be future-dated without changing historical transactions.

#### Dependencies

#### Show:

UAE VAT Profile is currently consumed by 1,284 products across 6 venues.

#### Acceptance Condition

TICVAI can maintain reusable, effective-dated tax profiles for all supported legal entities and jurisdictions.

## 10.3.3 — Tax Rule & Treatment Builder

### Purpose

Define how taxes are applied to products and transactions.

### Tax Treatments

#### Support:

- Tax Inclusive

- Tax Exclusive
- Tax Exempt
- Zero Rated
- Out of Scope

### Calculation Methods

#### Support:

- Percentage
- Fixed Tax
- Tiered
- Compound
- Sequential
- Multiple Concurrent Taxes

### Example — Inclusive

#### Displayed Price:

**AED 105**

#### VAT:

**AED 5**

#### Net:

**AED 100**

### Example — Exclusive

#### Base:

**AED 100**

#### VAT:

**AED 5**

#### Final:

**AED 105**

### Rule Conditions

#### Tax treatment can depend on:

- Product
- Product Category
- Venue
- Country
- Legal Entity
- Transaction Type
- Customer Type where legally relevant
- Sales Channel where legally relevant
- Effective Date

### Multiple Taxes

#### Example:

Base Price

→ Entertainment Tax

→ Municipality Fee

→ VAT

The sequence must be configurable.

### **Tax Exemption**

Rules may reference approved exemption conditions configured in Screen 10.3.6.

### **Acceptance Condition**

TICVAI correctly determines the applicable tax treatment and calculation method for every taxable commercial transaction.

---

## **10.3.4 — Fee & Surcharge Library**

### **Purpose**

Create standardized reusable non-base-price charges.

### **Fee Types**

#### **Support:**

- Booking Fee
- Transaction Fee
- Service Fee
- Convenience Fee
- Delivery Fee
- Handling Fee
- Modification Fee
- Rescheduling Fee
- Cancellation Fee
- Refund Fee
- Payment Fee
- Channel Fee
- Facility Fee
- Surcharge
- Custom Fee

### **Fee Configuration**

#### **Capture:**

- Fee Name
- Fee Code
- Fee Type
- Description
- Calculation Method
- Value
- Currency
- Tax Treatment
- Refundability

- Effective Period
- Status

### Calculation Methods

#### Support:

- Fixed Amount
- Percentage
- Per Ticket
- Per Product
- Per Person
- Per Order
- Per Transaction
- Per Day
- Tiered

### Example

#### Call Center Booking Fee

AED 15 / Order

#### Online Service Fee

3% / Transaction

### Fee Visibility

#### Configure whether the fee is:

- Customer Visible
- Included in Display Price
- Shown Separately
- Internal Only

### Reuse

One fee definition should be reusable across multiple fee rules.

### Acceptance Condition

TICVAI maintains a centralized reusable library of commercial fees and surcharges rather than defining them separately within sales channels.

---

## 10.3.5 — Fee Applicability & Charging Rule Builder

### Purpose

Determine **when a fee or surcharge should apply**.

Screen 10.3.4 defines the fee.

Screen 10.3.5 defines the conditions that trigger it.

### Conditions

#### Support:

- Product
- Product Category
- Channel

- Venue
- Event
- Customer Type
- Membership
- Transaction Type
- Payment Method
- Delivery Method
- Order Value
- Quantity
- Market
- Country
- Service Action

### Example

#### Call Center Fee

##### IF:

Channel = Call Center

##### THEN:

AED 15 per Order.

### Example

#### Modification Fee

##### IF:

Action = Reschedule

AND

Event occurs within 48 hours

##### THEN:

5% of affected ticket value.

### Threshold Rules

#### Example:

Order Value < AED 100

→ AED 5 handling fee.

Order Value ≥ AED 100

→ No handling fee.

### Fee Combination

#### Configure whether fees can:

- Stack
- Replace
- Exclude Another Fee
- Apply Once
- Apply Per Item

### Priority

#### Support:

- Rule Priority
- Stop Processing
- Continue Processing
- Mutual Exclusion

### **Conflict Detection**

#### **Example:**

Service Fee A and Service Fee B both apply to the same B2C transaction.

### **Acceptance Condition**

TICVAI applies fees only when their configured commercial and transactional conditions are satisfied.

---

## **10.3.6 — Fee Waiver, Tax Exemption & Exception Rules**

### **Purpose**

Govern circumstances under which a normally applicable tax or fee may be reduced, waived or exempted.

### **Exception Types**

#### **Support:**

- Fee Waiver
- Fee Reduction
- Tax Exemption
- Zero-Rated Tax
- Complimentary Transaction
- Operational Waiver
- Contractual Waiver

### **Eligibility Conditions**

#### **Configure by:**

- Membership Benefit
- Loyalty Tier
- Corporate Agreement
- B2B Contract
- Customer Segment
- Staff Role
- Promotion
- Service Recovery
- Operational Issue
- Legal Exemption
- Supervisor Override

### **Example**

#### **Gold Member**

#### **Booking Fee:**

#### **Waived**

**Example****Service Recovery****Supervisor may waive:****Rescheduling Fee**

Reason mandatory.

**Manual Override****Authorized staff can request:**

Waive AED 25 modification fee.

**System should require:**

- Reason
- User
- Permission
- Approval if configured
- Timestamp

Board 4 owns the broader approval framework.

**Tax Exemption Evidence****Where required, retain:**

- Exemption Type
- Reference
- Evidence
- Validity
- Verification Status

**Audit**

Every exemption/waiver must be traceable.

**Acceptance Condition**

Fees and taxes can only be waived or exempted through explicitly authorized rules or governed manual exceptions.

---

**10.3.7 — Price Calculation Sequence & Formula Engine****Purpose**

Define the exact sequence TICVAI follows to calculate the final payable amount.

This is the **heart of Board 3**.

**Recommended Calculation Pipeline****1. Commercial Base Rate**

↓

**2. Contextual Rate Selection**

↓

**3. Dynamic Pricing Adjustment**

↓

**4. Eligible Promotion / Discount**

↓

**5. Package / Bundle Adjustment**

↓

**6. Fees / Surcharges**

↓

**7. Tax Calculation**

↓

**8. Rounding**

↓

**9. Final Payable Amount**

The sequence must be configurable where business/legal requirements require differences.

**Calculation Components****Each step should define:**

- Input
- Formula
- Sequence
- Taxability
- Rounding
- Dependency
- Output

**Formula Builder****Support:**

- Fixed Amount
- Percentage
- Percentage of Base
- Percentage of Subtotal
- Tiered
- Conditional
- Minimum
- Maximum
- Custom Governed Formula

**Example****Selected Rate:**

AED 200

**Promotion:**

–AED 20

**Subtotal:**

AED 180

**Service Fee:**

+AED 10

**VAT:**

+AED 9.50

**Final:**

**AED 199.50**

**Formula Dependency**

TICVAI should prevent invalid circular formulas.

**Versioning**

Calculation formulas must be effective-dated so historical transactions remain reproducible.

**Acceptance Condition**

The system can deterministically calculate the final payable amount using a governed and traceable sequence of commercial components.

---

## 10.3.8 — Currency Precision, Rounding & Monetary Rules

**Purpose**

Ensure monetary calculations remain consistent across countries, currencies, channels and payment systems.

**Currency Rules**

**Configure:**

- Currency
- Decimal Places
- Minimum Monetary Unit
- Display Precision
- Calculation Precision
- Rounding Method

**Rounding Methods**

**Support:**

- Standard
- Round Up
- Round Down
- Banker's Rounding
- Nearest Currency Unit
- Custom Regulatory Rule

**Example**

**Calculated:**

AED 199.497

**Display:**

**AED 199.50**

**Cash Rounding**

**Where applicable:**

**Calculated Total:**

CHF 19.98

**Cash Payable:**

CHF 20.00

while electronic payment may remain CHF 19.98.

### **Rounding Stage**

#### **Configure whether rounding occurs:**

- Per Item
- Per Tax
- Per Fee
- Per Line
- At Order Total

### **Reconciliation Protection**

#### **The engine should ensure:**

Line totals + tax + fees = final transaction total.

### **Currency Conversion**

FX and multi-currency configuration may provide inputs, but Board 3 controls how the resulting monetary values are rounded/calculated.

### **Acceptance Condition**

TICVAI produces mathematically consistent monetary totals according to configured currency precision and rounding rules.

---

## **10.3.9 — Price Breakdown, Calculation Simulation & Explainability**

### **Purpose**

Allow administrators to test the complete calculation before releasing configuration into production.

### **Simulation Inputs**

#### **Select:**

- Customer
- Product
- Quantity
- Venue
- Event
- Channel
- Date
- Timeslot
- Membership
- Promotion
- Payment Method
- Delivery Method
- Currency

### **Example Result**

#### **Adult Admission × 2**

Selected Rate

AED 250 × 2 = AED 500

Member Adjustment

– AED 50

Promotion

– AED 25

Booking Fee

- AED 10

Subtotal

AED 435

VAT

- AED 21.75

**Final Payable**

**AED 456.75**

**Explainability Panel**

**For every component show:**

- Source
- Rule
- Formula
- Input
- Output
- Reason
- Tax Treatment
- Sequence

**Rule Trace**

**Example:**

Gold Member Rate selected by Pricing Rule 10.2.3.

B2C Service Fee applied by Fee Rule FE-021.

UAE VAT 5% applied using TAX-UAE-01.

**Channel Comparison**

**Run the same transaction through:**

**B2C / POS / Kiosk / B2B / Call Center**

to verify consistent calculations where expected.

**AI QA**

**AI can highlight:**

Call Center total differs from B2C by AED 15 because Call Center Booking Fee applies.

This is expected, not an error.

**Acceptance Condition**

Authorized users can reproduce and explain every monetary component contributing to the final payable amount before configuration goes live.

## 10.3.10 — Calculation Validation, Reconciliation & Service Interface

### Purpose

Provide final technical and commercial validation of the pricing calculation engine and define how other TICVAI modules consume it.

### Validation Areas

#### Check:

#### Tax

- Missing Profile
- Invalid Rate
- Expired Rule
- Overlapping Rule

#### Fees

- Duplicate Fee
- Conflicting Rule
- Missing Tax Treatment

#### Formula

- Circular Dependency
- Invalid Sequence
- Missing Input

#### Currency

- Invalid Precision
- Unsupported Currency
- Rounding Difference

#### Reconciliation

- Line Total
- Tax Total
- Fee Total
- Discount Total
- Final Total

#### Test Suite

#### Administrators should maintain test scenarios such as:

- Standard B2C Sale
- POS Sale
- Member Sale
- Group Booking
- B2B Sale
- Refund
- Reschedule
- Multi-Product Order
- Package Sale

- Multi-Currency Sale

## Regression Testing

### Before major calculation changes:

Run 500 predefined pricing scenarios.

### Expected:

**498 Passed**

**2 Failed**

The configuration should not progress until critical failures are resolved.

## Central Calculation Service

Other TICVAI modules should send transaction context to the central engine and receive:

- Selected Price
- Discounts
- Fees
- Taxes
- Rounding
- Final Payable
- Calculation Version
- Explanation Reference

## Example

### Input

Product + Customer + Channel + Quantity + Date + Context

↓

### TICVAI Pricing Engine

↓

### Output

Base: AED 250

Discount: AED 25

Fee: AED 10

VAT: AED 11.75

Final: **AED 246.75**

## API Responsibility

Board 3 owns the **calculation service and commercial response**.

### The central API/Developer Management module owns:

- Authentication
- API Keys
- Rate Limits
- Technical API Lifecycle
- Developer Documentation

## Historical Reproducibility

Given an old transaction and its calculation version, TICVAI must be able to explain how its original total was produced.

## Acceptance Condition

The calculation engine passes configured reconciliation and regression tests and exposes one authoritative calculation service to all TICVAI sales and operational modules.

## Board 3 — Final Screen Register

| #       | Backend Screen                                             | Core Responsibility                |
|---------|------------------------------------------------------------|------------------------------------|
| 10.3.1  | Tax, Fee & Calculation Command Center                      | Calculation configuration health   |
| 10.3.2  | Tax Profile & Jurisdiction Configuration                   | Tax master configuration           |
| 10.3.3  | Tax Rule & Treatment Builder                               | Tax applicability/calculation      |
| 10.3.4  | Fee & Surcharge Library                                    | Reusable fee definitions           |
| 10.3.5  | Fee Applicability & Charging Rule Builder                  | Fee conditions                     |
| 10.3.6  | Fee Waiver, Tax Exemption & Exception Rules                | Exemptions and waivers             |
| 10.3.7  | Price Calculation Sequence & Formula Engine                | Final calculation logic            |
| 10.3.8  | Currency Precision, Rounding & Monetary Rules              | Monetary accuracy                  |
| 10.3.9  | Price Breakdown, Calculation Simulation & Explainability   | Testing and transparency           |
| 10.3.10 | Calculation Validation, Reconciliation & Service Interface | QA and central calculation service |

### Area 10 architecture after Board 3

We now have a very clean separation:

## Board 1 — Commercial Pricing Foundation

### What prices exist?

↓

## Board 2 — Pricing Rules & Eligibility

### Which price applies?

↓

## **Board 3 — Tax, Fees & Calculation Engine**

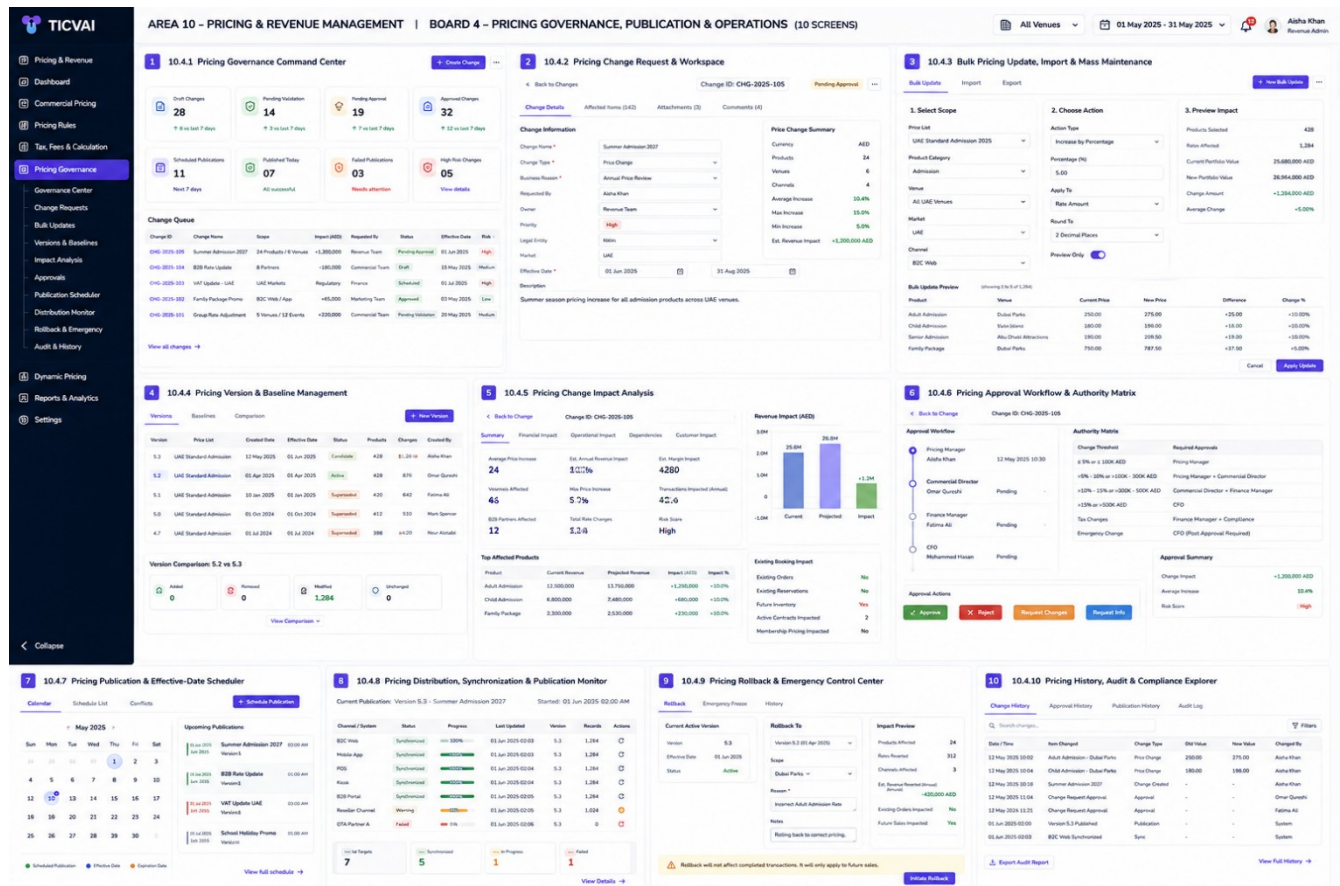
### **How do we reach the final payable amount?**

↓

**Board 4 — Pricing Governance, Publication & Operations**  
**Who can change pricing, approve it, version it and publish it?**

## Area 10 — Pricing &amp; Revenue Management

## Board 4 — Pricing Governance, Publication &amp; Operations



## 10 Backend Configuration Screens

## Board Objective

Boards 1-3 established the commercial and calculation engines:

**Board 1:** What prices exist?

**Board 2:** Which price applies?

**Board 3:** How is the final payable amount calculated?

**Board 4 governs:**

How pricing changes are created, reviewed, approved, versioned, scheduled, published, monitored, rolled back, and audited.

This is the **control and governance layer** protecting TICVAI from unauthorized or accidental commercial changes.

**The lifecycle should be:**

**Change Request → Impact Analysis → Bulk/Individual Change → Validation → Approval → Schedule → Publish → Synchronize → Monitor → Rollback/Audit**

**A critical principle:**

**No pricing configuration should become commercially effective without passing the applicable governance policy.**

## 10.4.1 — Pricing Governance Command Center

### Purpose

Provide Commercial, Revenue, Finance, and authorized management with one operational view of all pricing changes and governance activities.

### KPI Cards

#### Display:

- Pricing Changes in Draft
- Pending Validation
- Pending Approval
- Approved Changes
- Scheduled Publications
- Published Today
- Failed Publications
- Emergency Changes
- Rollbacks
- Expiring Prices
- Governance Exceptions
- High-Risk Changes

### Change Queue

#### Display:

| Change                | Scope       | Impact     | Requested By | Status           |
|-----------------------|-------------|------------|--------------|------------------|
| Summer Admission 2027 | 24 Products | +AED 1.2M  | Revenue Team | Pending Approval |
| B2B Rate Update       | 8 Partners  | −AED 180K  | Commercial   | Draft            |
| VAT Update            | UAE         | Regulatory | Finance      | Scheduled        |

### Filters

#### Filter by:

- Price List
- Product
- Venue
- Market
- Channel
- Legal Entity
- Change Type
- Owner
- Approver

- Risk
- Status
- Effective Date

### Governance Alerts

#### Examples:

**7 pricing changes are scheduled to activate within the next 24 hours.**

**2 changes exceed the configured 15% commercial-change threshold.**

**One publication failed synchronization to the B2B channel.**

#### Quick Actions

- Create Pricing Change
- Bulk Update
- Import Changes
- Review Impact
- Approve
- Schedule Publication
- View Rollbacks

### AI Governance Assistant

#### AI can summarize:

14 pending changes affect 286 products. Three are classified High Risk because projected revenue impact exceeds AED 500K.

#### Acceptance Condition

Authorized administrators can monitor and control the complete pricing-change lifecycle from a centralized governance workspace.

---

## 10.4.2 — Pricing Change Request & Workspace

### Purpose

Provide a governed workspace for creating individual or structured pricing changes before modifying production pricing.

### Change Request

#### Capture:

- Change ID
- Change Name
- Change Type
- Business Reason
- Requested By
- Owner
- Business Unit
- Legal Entity
- Market
- Venue
- Effective Date

- Expiry Date
- Priority
- Supporting Notes
- Attachments

## Change Types

### Support:

- Price Change
- New Rate
- Rate Removal
- Price List Change
- Eligibility Rule Change
- Tax Change
- Fee Change
- Formula Change
- Currency/Rounding Change
- Emergency Change

## Before / After

### Example:

#### Adult Admission

#### Current:

AED 250

#### Proposed:

AED 275

#### Difference:

**+AED 25 / +10%**

## Multi-Object Change

### One request may contain:

- Multiple Products
- Multiple Rates
- Multiple Venues
- Multiple Markets

## Change Reason

### Standard reasons:

- Annual Price Review
- New Season
- Commercial Strategy
- Contract Update
- Regulatory Change
- Cost Increase
- Market Adjustment
- Correction

- Emergency
- Other

### **Draft Workspace**

Changes remain isolated from live pricing until approved and published.

### **AI Assistance**

AI may summarize the proposed change and identify potentially unusual modifications.

### **Acceptance Condition**

All governed pricing modifications can originate from a traceable change request without directly editing live commercial configuration.

---

## **10.4.3 — Bulk Pricing Update, Import & Mass Maintenance**

### **Purpose**

Allow large pricing portfolios to be updated efficiently without manually editing hundreds or thousands of records.

### **Bulk Selection**

#### **Select by:**

- Price List
- Product
- Product Family
- Category
- Venue
- Market
- Currency
- Rate Type
- Channel
- Effective Period

### **Bulk Operations**

#### **Support:**

- Increase by %
- Decrease by %
- Increase Fixed Amount
- Decrease Fixed Amount
- Replace Amount
- Copy Rate
- Change Currency
- Apply Rounding
- Set Effective Dates
- Activate/Deactivate
- Clone for New Season

### **Example**

#### **Select:**

## 428 Admission Rates

### Action:

**Increase 5%**

### Preview:

Current portfolio value → Proposed portfolio value.

### Spreadsheet Import

#### Support controlled import of:

- CSV/XLSX pricing files
- Price lists
- Rate values
- Effective dates
- Mappings

### Import Validation

#### Before committing:

- Invalid Product
- Unknown Rate Code
- Duplicate Record
- Unsupported Currency
- Missing Mandatory Field
- Invalid Date
- Invalid Amount

### Preview

#### Show:

**Records Read:** 5,420

**Valid:** 5,371

**Warnings:** 37

**Errors:** 12

Nothing should change until validation succeeds.

### Export

Allow authorized users to export current pricing for controlled offline review.

### AI Mapping

#### AI may assist in mapping imported columns:

“Ticket Price” appears to correspond to Rate Amount.

But the administrator confirms mappings.

### Acceptance Condition

Large-scale pricing changes can be safely prepared, validated, previewed, and submitted through controlled bulk operations.

---

## 10.4.4 — Pricing Version & Baseline Management

### Purpose

Maintain immutable versions of pricing configuration so TICVAI always knows what configuration existed at a particular time.

### Version Register

#### Display:

- Version
- Price List
- Created Date
- Effective Date
- Created By
- Change Request
- Product Count
- Change Count
- Status

### Version States

**Draft / Candidate / Approved / Scheduled / Active / Superseded / Rolled Back / Archived**

### Version Comparison

#### Compare:

#### Version 4.2

vs.

#### Version 4.3

#### Show:

- Added
- Removed
- Modified
- Unchanged

### Difference Viewer

#### Example:

| Rate   | V4.2    | V4.3    | Difference |
|--------|---------|---------|------------|
| Adult  | AED 250 | AED 275 | +10%       |
| Child  | AED 180 | AED 195 | +8.3%      |
| Senior | AED 190 | AED 200 | +5.3%      |

### Baseline

**Allow administrators to designate an approved configuration as:**

### Commercial Baseline

Future changes can be compared against it.

### Historical Integrity

Historical orders must reference the exact pricing/calculation version used when the transaction occurred.

### Acceptance Condition

TICVAI maintains immutable, comparable pricing versions and can identify exactly which configuration was effective for any historical transaction.

---

## 10.4.5 — Pricing Change Impact Analysis

### Purpose

Determine the commercial and operational consequences of a pricing change **before approval and publication**.

This is one of the most important screens in Board 4.

### Impact Dimensions

#### Analyze:

- Products
- Events
- Performances
- Venues
- Markets
- Channels
- B2B Partners
- Memberships
- Packages
- Existing Reservations
- Future Reservations
- APIs
- Integrations

### Commercial Impact

#### Estimate:

- Current Revenue
- Projected Revenue
- Average Price Change
- Maximum Change
- Minimum Change
- Margin Impact
- Customer Exposure
- Transaction Volume

### Example

#### Proposed change:

**Adult Admission AED 250 → AED 275****Affected:**

42 Products  
 6 Venues  
 4 Channels  
 18 B2B Agreements

**Estimated annual revenue impact:****+AED 3.4M****Dependency Impact****Show whether changes affect:**

- Packages
- Derived Rates
- Contract Rates
- Membership Rates
- Group Rates
- Promotions
- Dynamic Pricing Guardrails

**Existing Booking Protection****Clearly identify whether the proposed change affects:****Existing Orders:** No**Existing Reservations:** No**Future Unsold Inventory:** Yes**Risk Score****Classify:****Low / Medium / High / Critical**

based on configurable criteria.

**AI Impact Summary****Example:**

The proposed increase affects 71% of weekend admission inventory. Gold Member rates remain unchanged, increasing the member discount differential from 20% to 27%.

**Acceptance Condition**

Approvers can understand financial, customer, channel, dependency, and operational impact before authorizing a pricing change.

## 10.4.6 — Pricing Approval Workflow & Authority Matrix

**Purpose**

Configure who must approve pricing changes based on their commercial risk and scope.

**Approval Rules****Approval can depend on:**

- Change %
- Monetary Impact

- Revenue Impact
- Price List
- Product
- Venue
- Market
- Legal Entity
- Channel
- Tax Change
- Fee Change
- Emergency Status

### Example Authority Matrix

| Change     | Approval                      |
|------------|-------------------------------|
| ≤5%        | Pricing Manager               |
| >5-10%     | Commercial Director           |
| >10-15%    | Commercial Director + Finance |
| >15%       | CFO / Authorized Executive    |
| Tax Change | Finance + Compliance          |

### Multi-Level Workflow

#### Example:

#### Pricing Manager

↓

#### Commercial Director

↓

#### Finance

↓

#### Final Approval

#### Approval Actions

- Approve
- Reject
- Return for Modification
- Request Information
- Delegate
- Escalate

### Segregation of Duties

#### Support policies such as:

User who created the pricing change cannot provide final approval.

**Approval SLA****Configure:**

- Expected Approval Time
- Escalation
- Reminder
- Alternate Approver

**AI Approval Summary****AI can summarize:**

Proposed change increases 84 rates by an average of 8.4%. Forecast revenue impact is +AED 1.1M. No tax or contractual pricing conflicts were detected.

AI does **not** approve the change.

**Audit****Every approval action records:**

- User
- Role
- Date/Time
- Decision
- Comment
- Version

**Acceptance Condition**

Pricing changes cannot progress beyond their configured governance threshold without all required approvals.

---

**10.4.7 — Pricing Publication & Effective-Date Scheduler****Purpose**

Control exactly **when approved pricing becomes commercially effective**.

**Publication Modes****Support:**

- Publish Immediately
- Scheduled Publication
- Future Effective Date
- Staged Publication
- Market-by-Market
- Venue-by-Venue
- Channel-by-Channel

**Scheduler****Calendar/timeline should show:**

- Publication Date
- Effective Date
- Expiry Date
- Venue

- Market
- Channel
- Version

**Example****Summer 2027 Pricing****Approved:**

15 February

**Publish Configuration:**

1 March 02:00

**Sales Effective:**

1 April 00:00

**Visit Effective:**

1 June

These dates may differ.

**Pre-Publication Check****Before publishing verify:**

- Approval Complete
- Validation Passed
- No Critical Conflicts
- Dependencies Available
- Channels Ready
- Effective Dates Valid

**Scheduled Changes****Show:**

**Next 7 Days / 30 Days / 90 Days**

**Collision Detection****Example:**

Version 5.2 and Version 5.3 are both scheduled to become effective for Dubai Adult Admission on 1 June.

**Cancellation**

Authorized users may cancel a scheduled publication before activation.

**Acceptance Condition**

Only validated and fully approved pricing configurations can be scheduled and activated at controlled effective dates.

## 10.4.8 — Pricing Distribution, Synchronization & Publication Monitor

**Purpose**

Ensure published pricing reaches every TICVAI channel and dependent system consistently.

**Distribution Targets****Monitor:**

- B2C
- Mobile App
- POS
- Mobile POS
- Kiosk
- Call Center
- B2B
- Reseller
- OTA
- APIs
- Cache/CDN where applicable
- External Integrated Systems

### **Publication Status**

#### **For each target:**

**Pending / Publishing / Synchronized / Warning / Failed**

#### **Example**

Version **2027.04**

B2C → Synchronized

POS → Synchronized

Kiosk → Synchronized

B2B → Synchronized

OTA A → Failed

OTA B → Synchronized

### **Failure Management**

#### **For failed publication:**

- Retry
- Queue
- Investigate
- Rollback Target
- Suspend Target

### **Consistency Check**

TICVAI should verify that the same governed pricing version is available across relevant channels.

### **Propagation Monitoring**

#### **Display:**

- Publication Started
- Last Updated
- Records Published
- Records Failed
- Latency
- Target Version

**Alerts**

OTA Partner X is still using pricing version 4.7 while current production version is 4.8.

**Acceptance Condition**

Administrators can confirm that approved pricing has been successfully distributed and synchronized to all intended channels and systems.

---

**10.4.9 — Pricing Rollback & Emergency Control Center****Purpose**

Provide controlled recovery when a pricing publication is incorrect or creates unacceptable commercial impact.

**Rollback Options****Support:**

- Previous Version
- Selected Version
- Previous Price
- Commercial Baseline
- Selected Products
- Selected Venue
- Selected Market
- Selected Channel
- Entire Publication

**Example****Current:****Version 5.3****Issue:**

Incorrect Adult Rate.

**Rollback to:****Version 5.2****Scope:**

Dubai Venue only.

**Impact Preview****Before rollback show:**

- Products Affected
- Channels Affected
- Transactions Since Activation
- Revenue Exposure
- Existing Orders
- Future Sales

**Historical Transaction Protection**

Rollback must **never rewrite completed transactions**.

It only changes pricing for applicable future transactions.

**Emergency Freeze****Authorized users should have:**

- Freeze Price List
- Freeze Product Pricing
- Freeze Venue Pricing
- Stop Scheduled Publication
- Stop Distribution
- Restore Last Known Good Version

**Emergency Override****Emergency action requires:**

- Reason
- Authorized Role
- Incident Reference
- Timestamp

Depending on policy, retrospective approval may be required.

**AI Detection****AI may identify:**

Conversion dropped 43% within 20 minutes of Version 5.3 activation. Pricing anomaly detected.

It may recommend rollback but should not bypass configured authority.

**Acceptance Condition**

Authorized users can rapidly contain and reverse problematic pricing changes without altering historical transactions or losing audit traceability.

---

**10.4.10 — Pricing History, Audit & Compliance Explorer****Purpose**

Provide complete forensic traceability for every pricing configuration and change.

**Audit Search****Search by:**

- Price List
- Rate
- Product
- User
- Change Request
- Version
- Venue
- Market
- Channel
- Date
- Transaction
- Approval

## Complete Change Record

### Store:

#### Old Value → New Value

### plus:

- Changed By
- Date/Time
- Reason
- Source
- Change Request
- Approval
- Version
- Effective Date
- Publication
- Rollback if applicable

### Example Timeline

**10:02** — Pricing Manager proposed AED 250 → AED 275

**10:18** — Impact analysis completed

**11:04** — Commercial Director approved

**11:21** — Finance approved

**02:00 next day** — Version 5.3 published

**02:03** — All channels synchronized

### Audit Types

#### Maintain:

- Configuration Audit
- Approval Audit
- Publication Audit
- Synchronization Audit
- Override Audit
- Rollback Audit
- Emergency Action Audit

### Compliance Export

#### Authorized users can export audit evidence for:

- Internal Audit
- Finance
- Compliance
- Regulatory Review

### Transaction Trace

#### From an order, administrators should be able to trace:

Which price list → which rule → which calculation profile → which version → which approvals produced this price?

### AI Audit Search

**Authorized users can ask:**

“Show all Adult ticket price changes above 10% during the last six months.”

**or:**

“Who approved the Dubai summer pricing?”

**Acceptance Condition**

Every material pricing configuration, approval, publication, override, and rollback is fully traceable and can be reconstructed for operational, financial, and compliance purposes.

## Board 4 — Final Screen Register

| #       | Backend Screen                                              | Core Responsibility            |
|---------|-------------------------------------------------------------|--------------------------------|
| 10.4.1  | Pricing Governance Command Center                           | Governance operations          |
| 10.4.2  | Pricing Change Request & Workspace                          | Controlled change creation     |
| 10.4.3  | Bulk Pricing Update, Import & Mass Maintenance              | Mass pricing operations        |
| 10.4.4  | Pricing Version & Baseline Management                       | Version control                |
| 10.4.5  | Pricing Change Impact Analysis                              | Pre-change impact              |
| 10.4.6  | Pricing Approval Workflow & Authority Matrix                | Approval governance            |
| 10.4.7  | Pricing Publication & Effective-Date Scheduler              | Controlled activation          |
| 10.4.8  | Pricing Distribution, Synchronization & Publication Monitor | Channel synchronization        |
| 10.4.9  | Pricing Rollback & Emergency Control Center                 | Recovery and emergency control |
| 10.4.10 | Pricing History, Audit & Compliance Explorer                | Full traceability              |

The pricing lifecycle is now very clear

## Board 1 — Commercial Pricing Foundation

### *What prices exist?*

↓

## Board 2 — Pricing Rules & Eligibility

### *Which price applies?*



## Board 3 — Tax, Fees & Calculation Engine

### *What is the final payable amount?*



## Board 4 — Pricing Governance, Publication & Operations

*Who can change it, who approves it, when does it become live, and how do we control it?*



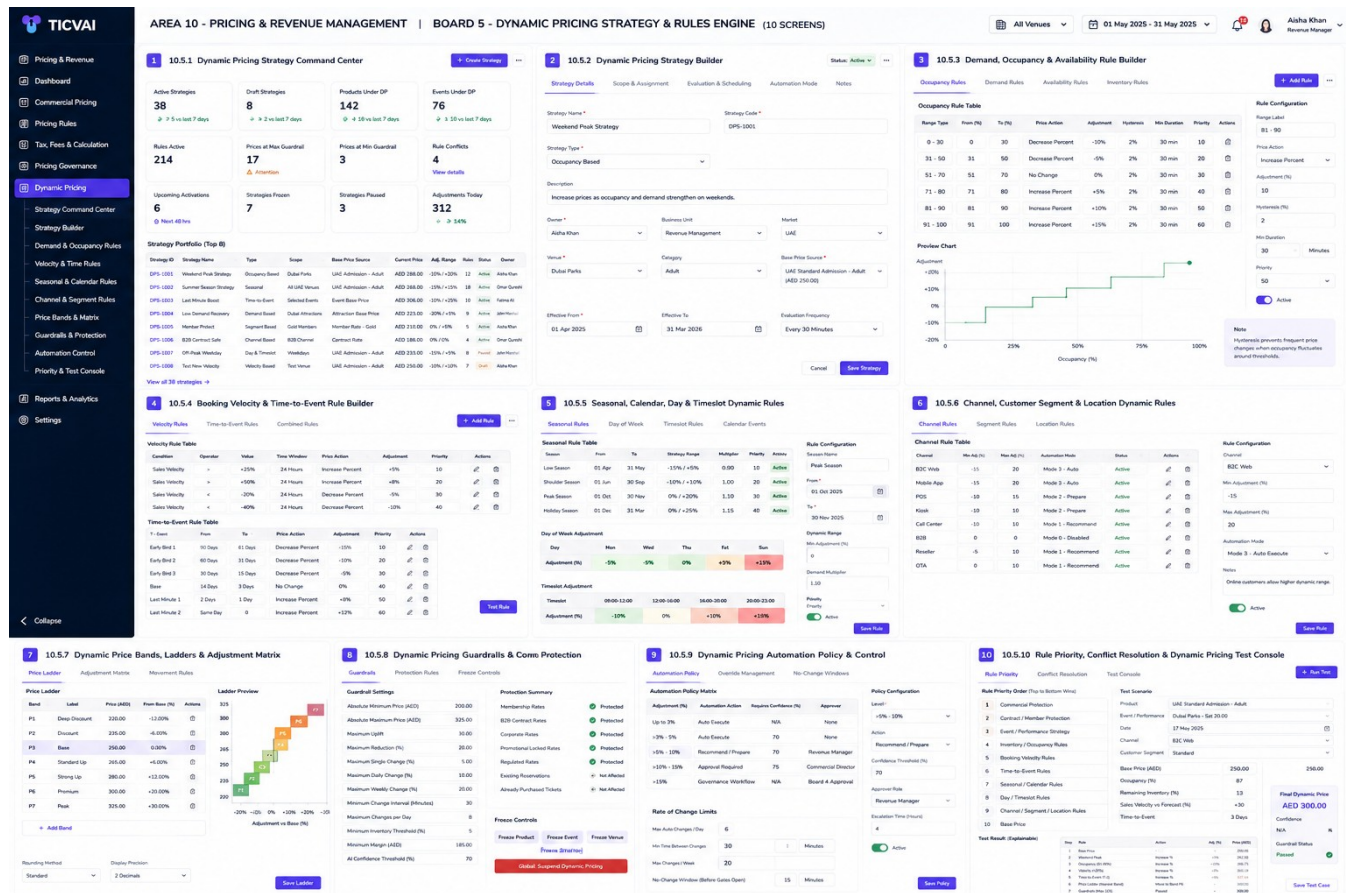
## Board 5 — Dynamic Pricing, Revenue Optimization & AI

### *How should TICVAI intelligently optimize the price?*

One architectural point I recommend we **freeze**: Board 4 should govern **all pricing changes**, including changes originating from AI/Dynamic Pricing when they exceed pre-approved automation guardrails. This gives Development team one governance engine rather than creating a separate approval mechanism inside Dynamic Pricing.

## Area 10 — Pricing &amp; Revenue Management

## Board 5 — Dynamic Pricing Strategy &amp; Rules Engine



## 10 Backend Configuration Screens — Revised Final SOW

Yes. I recommend we **replace the previous Board 5** entirely with this version.

Board 5 is now the **deterministic dynamic-pricing configuration engine**. It defines *how dynamic pricing behaves*. It does **not** own weather intelligence, nearby exhibitions/events, competitor intelligence, predictive AI, or advanced simulations; those will be handled cleanly in Boards 6 and 7.

## Board 5 Objective

Enable TICVAI revenue administrators to configure dynamic-pricing strategies using real-time commercial conditions such as:

**Demand + Occupancy + Availability + Inventory + Booking Velocity + Time-to-Event + Season + Day + Timeslot + Channel + Customer Segment + Location**

The engine converts those conditions into controlled price movements while always respecting commercial guardrails.

**The basic architecture is:**

**Base Commercial Price → Dynamic Context → Rules → Priority → Adjustment → Guardrails → Resolved Dynamic Price**

## 10.5.1 — Dynamic Pricing Strategy Command Center

### Purpose

Provide the central backend workspace for creating, monitoring, and managing all dynamic-pricing strategies.

This should be the primary operational screen for Revenue Managers.

### KPI Cards

#### Display:

- Active Strategies
- Draft Strategies
- Products Under Dynamic Pricing
- Events Under Dynamic Pricing
- Performances Under Dynamic Pricing
- Rules Active
- Current Price Adjustments
- Prices at Maximum Guardrail
- Prices at Minimum Guardrail
- Rule Conflicts
- Frozen Strategies
- Upcoming Activations

### Strategy Directory

#### Each strategy should show:

- Strategy ID
- Strategy Name
- Strategy Type
- Product/Event
- Venue
- Base Price Source
- Current Price
- Adjustment Range
- Rule Count
- Effective Period
- Automation Mode
- Status
- Owner

### Strategy Types

#### Support:

- Demand Based
- Occupancy Based
- Availability Based
- Inventory Based
- Booking Velocity

- Time-to-Event
- Seasonal
- Day-of-Week
- Timeslot
- Channel
- Segment
- Location
- Hybrid

### Status

**Draft / Testing / Ready / Scheduled / Active / Paused / Frozen / Expired / Retired**

### Quick Actions

- Create Strategy
- Duplicate
- Open
- Test
- Compare
- Pause
- Freeze
- Retire

### Operational Alerts

#### Examples:

12 performances reached their configured upper price band.

3 strategies contain unresolved rule conflicts.

Summer Weekend Strategy activates in 48 hours.

### Acceptance Condition

Authorized users can view and manage the complete dynamic-pricing strategy portfolio from one centralized workspace.

---

## 10.5.2 — Dynamic Pricing Strategy Builder

### Purpose

Create the master dynamic-pricing strategy and define what commercial objects it controls.

### Strategy Configuration

#### Configure:

- Strategy Name
- Strategy Code
- Description
- Strategy Type
- Owner
- Business Unit
- Market

- Venue
- Product
- Event
- Performance
- Timeslot
- Price Category
- Base Price Source
- Effective From
- Effective To
- Evaluation Frequency
- Status

### Base Price Source

The strategy must reference Board 1 pricing.

#### Example:

UAE Standard Admission → Adult → AED 250

Dynamic Pricing should **never create an uncontrolled independent base price.**

### Scope Assignment

#### One strategy may apply to:

- Single Product
- Product Family
- Event
- Multiple Performances
- Venue
- Selected Timeslots
- Selected Price Categories

### Evaluation Frequency

#### Examples:

- Every 15 Minutes
- Every 30 Minutes
- Hourly
- Daily
- On Inventory Change
- On Threshold Trigger

Actual technical frequency should remain configurable according to implementation and scale.

### Strategy Combination

#### Configure whether a strategy:

- Operates Independently
- Can Combine with Other Strategies
- Has Exclusive Control
- Acts as Fallback

**Clone****Example:**

Dubai Weekend Strategy

**can become:**

Abu Dhabi Weekend Strategy

with inherited rules but different commercial parameters.

**Acceptance Condition**

Administrators can create reusable dynamic-pricing strategies linked to governed commercial base prices and clearly defined product/event scopes.

**10.5.3 — Demand, Occupancy & Availability Rule Builder****Purpose**

Configure price movements driven by actual demand and capacity consumption.

**This directly covers the fundamental matrix requirements for:**

- Demand-Based Pricing
- Occupancy-Based Pricing
- Availability-Based Pricing
- Inventory-Based Pricing

**Supported Inputs****Rules can evaluate:**

- Tickets Sold
- Current Demand
- Occupancy %
- Remaining Capacity
- Remaining Inventory
- Available Seats
- Capacity Utilization
- Sales Pace

**Occupancy Matrix****Example:**

| Occupancy | Price Action |
|-----------|--------------|
| 0-30%     | -10%         |
| 31-50%    | -5%          |
| 51-70%    | Base         |
| 71-80%    | +5%          |
| 81-90%    | +10%         |

| Occupancy | Price Action |
|-----------|--------------|
| >90%      | +15%         |

## Inventory Rules

### Example:

#### IF

Remaining Inventory < 100

#### THEN

Increase Price +8%.

## Availability Rules

### Example:

#### IF

Available Capacity < 15%

#### THEN

Move to Price Band 4.

## Demand Rules

### Example:

#### IF

Demand Index > 1.25

#### THEN

Increase by 5%.

The definition of Demand Index must be explicitly configured and documented.

## Threshold Behavior

### Configure:

- Enter Threshold
- Exit Threshold
- Minimum Duration
- Cooldown Period

This prevents price bouncing.

### Example:

Occupancy crosses 80% → price increases.

If occupancy temporarily falls to 79.8%, TICVAI should not immediately reverse it unless configured.

## Acceptance Condition

TICVAI can automatically resolve governed price adjustments according to demand, occupancy, availability, and inventory thresholds.

## 10.5.4 — Booking Velocity & Time-to-Event Rule Builder

### Purpose

Control price movement based on **how quickly inventory is selling** and **how much time remains before the event or visit date**.

This is critical because occupancy alone is insufficient for effective revenue management.

### Booking Velocity Inputs

#### Support:

- Sales per Hour
- Sales per Day
- Sales per Week
- Current Booking Pace
- Expected Booking Pace
- Historical Booking Curve
- Pace Variance %
- Remaining Inventory

### Example

#### Expected:

100 tickets/day

#### Actual:

145 tickets/day

#### Velocity:

**+45%**

#### Rule:

Increase current price by 5%.

### Time-to-Event

#### Support:

- T-180
- T-90
- T-60
- T-30
- T-14
- T-7
- T-3
- T-1
- Same Day

or custom intervals.

### Early-Bird Rules

#### Example:

T-90 → -15%

T-60 → -10%

T-30 → -5%

T-14 → Base Price

### **Last-Minute Rules**

#### **Example:**

#### **IF**

T-2 Days

AND Occupancy <50%

THEN -10%.

#### **But:**

#### **IF**

T-2 Days

AND Occupancy >90%

THEN +15%.

### **Combined Rule**

#### **Example:**

IF booking velocity > forecast by 25% AND T-7 days AND occupancy >75%, increase 8%.

### **Price Decay / Escalation**

Allow progressive pricing movements as event time approaches.

### **Acceptance Condition**

Dynamic pricing can respond to both booking pace and remaining selling time rather than relying solely on current occupancy.

---

## **10.5.5 — Seasonal, Calendar, Day & Timeslot Dynamic Rules**

### **Purpose**

Configure dynamic pricing behavior according to temporal commercial patterns.

### **Supported Dimensions**

- Season
- Month
- Week
- Date Range
- Public Holiday
- School Holiday
- Day of Week
- Weekend
- Time of Day
- Timeslot
- Performance
- Special Date

### **Seasonal Strategy**

#### **Example:**

#### **Low Season**

**Dynamic Range:**

–15% → +5%

**Peak Season****Dynamic Range:**

Base → +20%

**Day-of-Week****Example:****Monday-Thursday:**

–5% to +10%

**Friday:**

Base to +15%

**Saturday/Sunday:**

Base to +20%

**Timeslot Rules****Example:**

09:00–12:00 → Off-Peak Strategy

12:00–16:00 → Standard Strategy

16:00–20:00 → Peak Strategy

**Special Calendar****Administrators can maintain:**

- Ramadan
- Eid
- Christmas
- New Year
- School Break
- National Day
- Custom Commercial Calendar

**Calendar Visualization**

Show dynamic pricing strategy coverage visually across dates and timeslots.

**Overlap Detection****Example:**

Eid Peak Rule overlaps with Weekend Rule and Special Event Rule.

System should identify which rule wins through Screen 10.5.10.

**Acceptance Condition**

TICVAI can dynamically adjust pricing according to configured seasons, dates, days, performances, and timeslots without creating conflicting temporal rules.

## 10.5.6 — Channel, Customer Segment & Location Dynamic Rules

### Purpose

Allow dynamic-pricing behavior to differ according to commercial context.

### Channel Dimensions

#### Support:

- B2C
- Mobile App
- POS
- Kiosk
- Call Center
- B2B
- Reseller
- OTA
- API

### Example

#### B2C:

Dynamic range  $\pm 15\%$

#### B2B Contract:

Dynamic Pricing Disabled

#### OTA:

Dynamic range +0% to +10%

### Customer Segment

#### Configure whether dynamic pricing applies to:

- Standard Customer
- Member
- Loyalty Tier
- Resident
- VIP
- Corporate
- Group
- B2B
- Custom Segment

### Protected Segments

#### Example:

Gold Members always receive their protected membership rate and are excluded from dynamic adjustment.

### Location

#### Rules can vary by:

- Country
- Market

- Venue
- Attraction
- Zone
- Event Location

### Example

#### Dubai Venue:

Maximum dynamic uplift +20%.

#### Abu Dhabi Venue:

Maximum +15%.

### Contract Protection

#### Dynamic pricing must respect:

- Partner Contract Rates
- Corporate Agreements
- Membership Guarantees
- Fixed Group Rates

### Acceptance Condition

Dynamic pricing can vary appropriately by channel, customer segment, and location without violating protected commercial agreements.

---

## 10.5.7 — Dynamic Price Bands, Ladders & Adjustment Matrix

### Purpose

Define the controlled monetary steps through which prices can move.

This is preferable to allowing unrestricted price generation for many products.

### Price Ladder Example

#### Base Price:

**AED 250**

#### Configured bands:

| Band | Price   |
|------|---------|
| P1   | AED 220 |
| P2   | AED 235 |
| P3   | AED 250 |
| P4   | AED 265 |
| P5   | AED 280 |

| Band | Price   |
|------|---------|
| P6   | AED 300 |
| P7   | AED 325 |

## Adjustment Models

### Support:

- Fixed Price Bands
- Percentage Bands
- Fixed Amount Steps
- Derived Bands
- Continuous Range where permitted

## Movement Rules

### Configure:

- Maximum Bands per Movement
- Minimum Time Between Movements
- Upward Movement
- Downward Movement
- Reversal Rules
- Cooldown Period

## Example

### Current:

P3 = AED 250

### Strong demand signal recommends:

AED 286

### If maximum movement is one band:

Resolved Price = P4 AED 265

rather than jumping directly to AED 286.

## Asymmetric Movement

### Support:

#### Increase:

Maximum +5% per adjustment

#### Decrease:

Maximum -10%

where required.

## Psychological Pricing

### Allow governed endpoints:

AED 249

AED 259

AED 279

rather than arbitrary decimals.

### Price Ladder Preview

**Show complete commercial movement from:**

**Minimum → Base → Maximum**

### Acceptance Condition

Dynamic prices move through controlled commercial price bands or adjustment ranges rather than generating arbitrary uncontrolled selling prices.

---

## 10.5.8 — Dynamic Pricing Guardrails & Commercial Protection

### Purpose

Establish the non-negotiable boundaries for every dynamic-pricing strategy.

### Core Guardrails

#### Configure:

- Absolute Minimum Price
- Absolute Maximum Price
- Minimum Margin
- Maximum Uplift %
- Maximum Reduction %
- Maximum Single Change
- Maximum Daily Change
- Maximum Weekly Change
- Minimum Change Interval
- Maximum Changes per Day
- Minimum Inventory
- Maximum Occupancy Trigger

### Example

#### Commercial Base:

AED 250

#### Minimum:

AED 200

#### Maximum:

AED 325

#### Dynamic engine calculates:

AED 347

#### Result:

**Capped at AED 325**

### Commercial Protection

#### Protect:

- Contract Rates

- Membership Rates
- Corporate Rates
- Promotional Locked Rates
- Regulatory Prices
- Complimentary Rates
- Already Purchased Tickets
- Existing Reservations where applicable

### Margin Protection

#### Example:

Never sell below AED 185 because minimum contribution margin would be breached.

### Extreme Movement Protection

#### Example:

Price cannot move more than 10% within 24 hours regardless of rule outcome.

### Freeze Controls

#### Authorized users can:

- Freeze Product
- Freeze Event
- Freeze Performance
- Freeze Strategy
- Freeze Venue
- Return to Base Price

### Global Kill Switch

#### Provide an authorized:

### Suspend Dynamic Pricing

control for critical incidents.

### Acceptance Condition

No dynamic pricing strategy can generate a selling price outside approved financial, contractual, operational, or regulatory boundaries.

## 10.5.9 — Dynamic Pricing Automation Policy & Control

### Purpose

Define **how much authority the pricing engine has to act on a calculated dynamic price**.

This is different from Board 4's approval workflow.

Board 5 determines **whether the engine may act automatically**.

Board 4 handles governance when formal approval is required.

### Automation Modes

#### Mode 0 — Monitor

Calculate but do not recommend or change.

#### Mode 1 — Recommend

Generate recommendation for Revenue Manager.

**Mode 2 — Prepare Change**

Create proposed pricing change automatically.

**Mode 3 — Auto-Execute Within Guardrails**

Automatically update price if all configured conditions are satisfied.

**Example Authority Policy**

| Adjustment | Action                                   |
|------------|------------------------------------------|
| ≤3%        | Auto-Execute                             |
| >3-5%      | Auto-Execute if confidence threshold met |
| >5-10%     | Revenue Manager                          |
| >10-15%    | Commercial Director                      |
| >15%       | Board 4 Governance                       |

AI confidence becomes relevant when Board 6 supplies an AI recommendation.

**Automation Scope****Configure independently by:**

- Product
- Event
- Venue
- Strategy
- Channel
- Market

**Rate of Change****Configure:**

- Maximum Automatic Changes / Day
- Minimum Time Between Changes
- No-Change Windows

**Example:**

Do not automatically change prices within 30 minutes of gates opening.

**Manual Override**

Authorized users can temporarily override the engine.

**Capture:**

- User
- Reason
- Override Price
- Start
- Expiry
- Return Behavior

**Safe Failure****If required pricing inputs are unavailable:****Hold Last Price / Return to Base / Freeze / Request Review****Acceptance Condition**

TICVAI provides configurable levels of pricing automation while routing changes outside authorized boundaries into the central governance process.

---

## 10.5.10 — Rule Priority, Conflict Resolution & Dynamic Pricing Test Console

**Purpose**

Determine the final dynamic price when multiple strategies and rules are simultaneously applicable.

This is the final and most important control screen of Board 5.

**Example****For one Saturday evening ticket:****Base Price:** AED 250**Applicable rules:**

Weekend → +5%

Occupancy 87% → +10%

Booking Velocity +30% → +5%

T-3 Days → +8%

Member Protection → Maximum AED 260

Maximum Commercial Price → AED 325

The engine needs a deterministic answer.

**Resolution Methods****Support:**

- Highest Priority Wins
- Most Specific Rule Wins
- Cumulative Adjustment
- Maximum Adjustment Wins
- Minimum Adjustment Wins
- Weighted Combination
- Stop Processing
- Custom Governed Resolution

**Priority Matrix**

Administrators configure rule hierarchy.

**Example:****Commercial Protection**

↓

**Contract/Member Protection**

↓

**Event-Specific Strategy**

↓

**Inventory/Occupancy**

↓

**Booking Velocity**

↓

**Time-to-Event**

↓

**Season/Day/Timeslot**

↓

**Base Price****Conflict Detection****Identify:**

- Contradictory Rules
- Same Priority
- Impossible Condition
- Overlapping Strategy
- Circular Dependency
- Missing Fallback
- Guardrail Conflict

**Dynamic Pricing Test Console****Inputs:**

- Product
- Event
- Performance
- Date
- Timeslot
- Channel
- Customer Segment
- Base Price
- Occupancy
- Inventory
- Booking Velocity
- Time-to-Event

**Example Result****Base Price**

AED 250

**Rules Matched**

Weekend +5%

Occupancy +10%

Velocity +5%

**Raw Calculated Price**

AED 300

**Price Ladder**

Nearest Allowed Band = AED 300

**Guardrail**

Passed

**Final Dynamic Price****AED 300****Explainability****Show:****Why AED 300?**

with the complete calculation path.

**Test Cases****Allow administrators to save regression cases:**

- Low Demand
- High Demand
- Near Sell-Out
- Early Bird
- Last Minute
- Weekend Peak
- Member Purchase
- B2B Contract
- Custom

**Acceptance Condition**

For any configured commercial scenario, TICVAI can deterministically calculate, test, and explain the final dynamic price before the strategy is activated.

## Board 5 — Final Screen Register

| #      | Backend Screen                                   | Primary Responsibility        |
|--------|--------------------------------------------------|-------------------------------|
| 10.5.1 | Dynamic Pricing Strategy Command Center          | Strategy portfolio            |
| 10.5.2 | Dynamic Pricing Strategy Builder                 | Strategy master configuration |
| 10.5.3 | Demand, Occupancy & Availability Rule Builder    | Demand/capacity rules         |
| 10.5.4 | Booking Velocity & Time-to-Event Rule Builder    | Pace/time rules               |
| 10.5.  | Seasonal, Calendar, Day & Timeslot Dynamic Rules | Temporal pricing              |

| #       | Backend Screen                                                    | Primary Responsibility     |
|---------|-------------------------------------------------------------------|----------------------------|
| 5       |                                                                   |                            |
| 10.5.6  | Channel, Customer Segment & Location Dynamic Rules                | Contextual dynamic pricing |
| 10.5.7  | Dynamic Price Bands, Ladders & Adjustment Matrix                  | Controlled price movement  |
| 10.5.8  | Dynamic Pricing Guardrails & Commercial Protection                | Commercial boundaries      |
| 10.5.9  | Dynamic Pricing Automation Policy & Control                       | Automation authority       |
| 10.5.10 | Rule Priority, Conflict Resolution & Dynamic Pricing Test Console | Resolution and testing     |

### Important Boundary for Development team

I would explicitly put this in the technical SOW because it prevents overlap later.

### Board 5 owns the Dynamic Pricing Engine

#### It consumes known commercial facts such as:

Occupancy = 87%

Booking velocity = +30%

Inventory remaining = 13%

T-3 days

Saturday evening

and determines **what pricing action should occur according to configured rules.**

### Board 6 will own the AI Intelligence & External Signal Layer

#### That is where we will properly configure:

**Weather intelligence → Nearby exhibition/event intelligence → Competitor pricing → Tourism/market demand → Public holidays → Historical pattern intelligence → AI forecasting → Elasticity → AI recommendations → Confidence and explainability.**

#### For example, Board 6 may tell Board 5:

#### Expected Demand Impact: +18%

Reason: major exhibition within 1.5 km + extreme outdoor temperature + booking acceleration

Confidence: 91%

Board 5 then determines what that intelligence is **allowed to do to the price.**

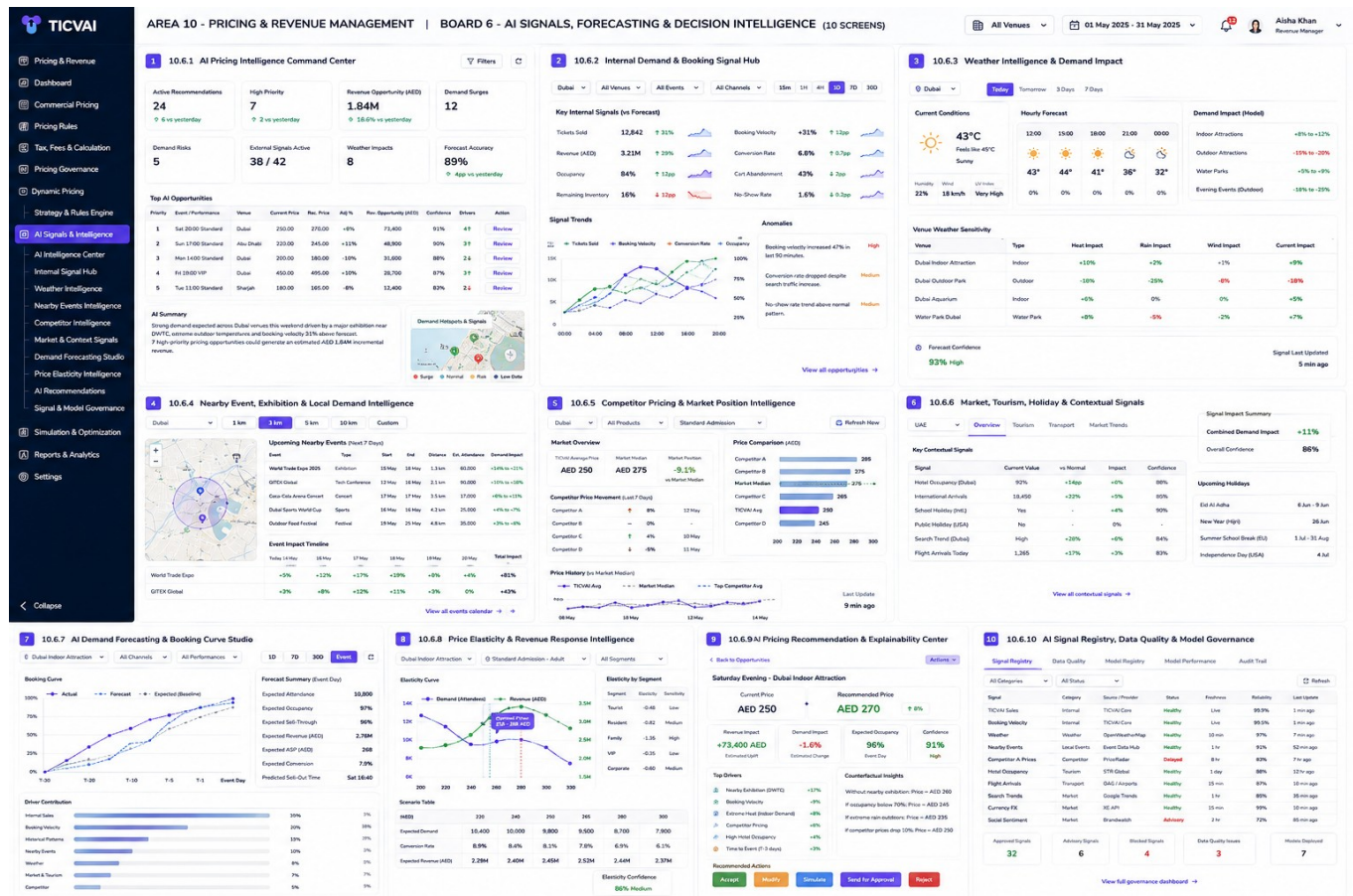
That separation is important: **AI predicts and recommends; the Dynamic Pricing Engine executes according to governed commercial rules.**

Then **Board 7** will handle the simulation, experimentation, live optimization, performance measurement and learning loop.

With this redesign, I consider **Board 5 properly scoped and ready to freeze** before we design its 10 screens.

## Area 10 — Pricing &amp; Revenue Management

## Board 6 — AI Signals, Forecasting &amp; Decision Intelligence



## 10 Backend Configuration Screens — Detailed SOW

Board 6 is where TICVAI's Dynamic Pricing becomes genuinely **AI-driven** rather than simply rule-driven.

## Board 5 answers:

“Given known conditions, what pricing rule should apply?”

## Board 6 answers:

“What is happening around the venue, what is likely to happen next, how will it affect demand, and what should TICVAI recommend?”

This board should combine **internal transactional signals** + **external contextual intelligence** + **forecasting/ML** + **price elasticity** + **explainable recommendations**.

## The architecture is:

**Internal Signals** + **Weather** + **Nearby Events** + **Competitors** + **Market/Tourism Signals** + **Historical Patterns**



**Signal Normalization & Quality**



**Demand Forecasting**



**Elasticity & Revenue Intelligence**

↓

**AI Pricing Recommendation**

↓

**Confidence + Explanation**

↓

**Board 5 Rules/Guardrails**

↓

**Board 7 Simulation/Execution****A fundamental requirement for Development team should be:**

**An LLM alone must not determine ticket prices.** AI recommendations should be based on governed structured data, statistical/ML forecasting, pricing rules and optimization models. Generative AI can provide interpretation, explanation and conversational interaction on top of those outputs.

---

## 10.6.1 — AI Pricing Intelligence Command Center

### Purpose

Provide Revenue Managers with a single operational view of all AI signals, forecasts, opportunities and risks influencing pricing.

### KPI Cards

#### Display:

- Active AI Recommendations
- High-Priority Opportunities
- Estimated Revenue Opportunity
- Demand Surges Detected
- Demand Risks Detected
- External Signals Active
- Nearby Events Detected
- Weather Impacts
- Competitor Movements
- Forecast Accuracy
- Average AI Confidence
- Data Quality Issues

### Opportunity Queue

#### Each opportunity displays:

- Product/Event
- Venue
- Current Price
- Recommended Price
- Adjustment %
- Demand Forecast
- Revenue Opportunity

- Primary Drivers
- Confidence
- Risk
- Urgency

#### Example:

#### Saturday Evening Admission

Current: AED 250

Recommended: AED 270

Expected Revenue Uplift: +AED 73,400

Confidence: 91%

#### Drivers

- ↑ Nearby Exhibition
- ↑ Booking Velocity
- ↑ Indoor Demand due to Weather
- ↑ Competitor Price
- ↓ Remaining Capacity

#### Geographic Intelligence

#### Map venues and nearby demand signals:

- Exhibitions
- Conferences
- Concerts
- Sports Events
- Festivals
- Tourism Events
- Weather Conditions

#### AI Summary

#### Example:

Dubai Venue demand is forecast 19% above baseline this weekend. The primary drivers are a major exhibition 1.3 km away, high hotel occupancy and stronger-than-expected booking velocity.

#### Drill-Down

#### Selecting any recommendation should expose:

**Signals → Forecast → Recommendation → Confidence → Expected Impact**

#### Acceptance Condition

Revenue teams can understand all significant AI pricing opportunities, risks, forecasts and external demand drivers from one centralized intelligence workspace.

## 10.6.2 — Internal Demand & Booking Signal Hub

### Purpose

Centralize the internal TICVAI signals used by forecasting and AI pricing models.

These are generally the highest-confidence signals because they come directly from TICVAI transactions.

### Signal Categories

**Support:****Sales**

- Tickets Sold
- Orders
- Revenue
- Average Selling Price
- Conversion Rate
- Cart Abandonment
- Search-to-Purchase Conversion

**Inventory**

- Capacity
- Remaining Inventory
- Availability
- Occupancy
- Seat/Zone Availability

**Velocity**

- Sales per Hour
- Sales per Day
- Booking Velocity
- Revenue Velocity
- Acceleration/Deceleration

**Customer**

- Segment
- Membership
- Geography
- Repeat Purchase
- Lead Time

**Operational**

- Cancellation
- Refund
- No-Show
- Reschedule

**Historical Comparison****Compare current performance against:**

- Yesterday
- Previous Week
- Same Day Last Year
- Previous Event
- Similar Event
- Forecast Baseline

**Signal Trend**

**Example:****Booking Velocity:****+31% vs forecast****Occupancy:****84%****Conversion:****+7.2%****Remaining Inventory:****16%****Anomaly Detection****AI should identify:**

Booking velocity increased 47% during the last 90 minutes.

**or:**

Conversion declined despite increased search traffic.

**Signal Granularity****Signals should be available by:**

**Tenant → Market → Venue → Event → Performance → Product → Price Category → Channel → Timeslot**

**Acceptance Condition**

TICVAI provides a governed, real-time internal signal layer for AI demand forecasting and pricing intelligence.

---

## 10.6.3 — Weather Intelligence & Demand Impact Configuration

**Purpose**

Allow TICVAI to understand how weather conditions affect demand for different venues and experiences.

This should be much more sophisticated than simply connecting a weather API.

**Weather Signals****Support:**

- Temperature
- Feels-Like Temperature
- Rain
- Rain Probability
- Humidity
- Wind
- Visibility
- Storm
- Extreme Heat
- Air Quality where available
- Current Conditions

- Hourly Forecast
- Daily Forecast

### **Venue Sensitivity**

#### **Configure venue characteristics:**

#### **Indoor / Outdoor / Mixed**

and weather sensitivity.

#### **Example:**

#### **Outdoor Theme Park:**

Extreme Heat → Negative

#### **Indoor Attraction:**

Extreme Heat → Potential Positive

#### **Water Park:**

High Temperature → Positive

#### **Outdoor Concert:**

Heavy Rain → Strong Negative

### **Weather Impact Model**

#### **Example:**

**Temperature >42°C**

#### **Indoor Venue:**

Demand Impact **+8%**

#### **Outdoor Venue:**

Demand Impact **−17%**

### **Forecast Horizon**

#### **Support:**

- Today
- 24 Hours
- 3 Days
- 7 Days
- Configurable Future Window

### **Historical Learning**

#### **The model should compare:**

Similar weather conditions historically resulted in +11% indoor demand.

### **Confidence**

#### **Show:**

Weather Forecast Confidence: **93%**

Estimated Demand Impact: **+8-12%**

### **Data Failure**

#### **If weather data becomes unavailable:**

- Use Last Valid Signal
- Use Historical Baseline
- Reduce Confidence

- Ignore Weather
- Suspend Weather-Driven Recommendation

### Acceptance Condition

Weather conditions are converted into venue-specific, explainable demand signals rather than directly changing prices.

---

## 10.6.4 — Nearby Event, Exhibition & Local Demand Intelligence

### Purpose

Detect external events around TICVAI venues that could materially affect visitor demand. This directly addresses the exhibition-near-the-venue scenario.

### Event Types

#### Detect/configure:

- Exhibition
- Conference
- Concert
- Sports Event
- Festival
- Trade Show
- Convention
- Public Celebration
- Major Attraction Event
- School Event
- Custom Local Event

### Event Data

#### Capture:

- Event Name
- Venue
- Location
- Distance from TICVAI Venue
- Start/End Date
- Start/End Time
- Expected Attendance
- Event Type
- Audience Type
- Source
- Confidence

### Geographic Radius

#### Configure per TICVAI venue:

Monitor events within **1 km / 3 km / 5 km / 10 km / Custom Radius**

**Demand Impact****Example:****Major Exhibition**

Distance: 1.2 km

Attendance: 60,000

Duration: 4 days

**Historical Correlation:**

+17% venue demand

**Predicted Impact:**

**+14-21%**

**Timing Intelligence****AI should understand that impact may occur:**

- Before Event
- During Event
- Lunch Period
- After Event
- Evening
- Following Day

**Audience Matching****Example:**

Family Exhibition + Family Attraction

→ High correlation.

Industrial Conference + Children's Attraction

→ Lower correlation.

**Event Calendar****Provide a calendar/map showing external events alongside:**

- TICVAI Events
- Demand Forecast
- Occupancy
- Current Pricing

**Acceptance Condition**

TICVAI can discover and evaluate nearby events and convert them into contextual demand-impact signals for AI pricing decisions.

---

## 10.6.5 — Competitor Pricing & Market Position Intelligence

**Purpose**

Allow TICVAI to understand its commercial position relative to relevant competitors.

**Competitor Registry****Configure:**

- Competitor
- Market

- Venue/Product
- Comparable TICVAI Product
- Source
- Currency
- Collection Method
- Refresh Frequency
- Reliability

## **Pricing Intelligence**

### **Track:**

- Published Price
- Promotional Price
- Weekend Price
- Peak Price
- Resident Price
- Member Price where publicly available
- Availability
- Date
- Timeslot

## **Comparison**

### **Example:**

#### **TICVAI Venue:**

AED 250

#### **Competitor A:**

AED 275

#### **Competitor B:**

AED 265

#### **Competitor C:**

AED 290

#### **Market Median:**

**AED 275**

#### **TICVAI Position:**

**–9.1% below median**

## **Competitor Movement Detection**

### **Example:**

Competitor A increased Saturday admission by 8%.

## **Comparable Product Mapping**

Administrators must define whether products are genuinely comparable.

### **This prevents AI comparing:**

General Admission

### **against:**

VIP Premium Package.

**Historical Correlation****Analyze whether competitor changes historically affected TICVAI:**

- Demand
- Conversion
- Price Sensitivity

**Ethical/Legal Controls**

Only approved and legally permissible data sources should be consumed.

**Acceptance Condition**

TICVAI can incorporate governed competitor and market-price intelligence without allowing competitor data alone to determine the selling price.

---

**10.6.6 — Market, Tourism, Holiday & Contextual Signal Hub****Purpose**

Capture broader external factors that may affect visitor demand beyond weather and nearby events.

**Signal Categories****Support:****Calendar**

- Public Holidays
- School Holidays
- Ramadan
- Eid
- Christmas
- New Year
- Long Weekends

**Tourism**

- Visitor Arrivals
- Tourism Demand
- Hotel Occupancy
- Hotel Rates
- Destination Demand

**Transport****Where data is available:**

- Flight Arrivals
- Airport Passenger Volume
- Public Transport Demand
- Traffic Conditions

**Market**

- Consumer Demand Trends
- Search Trends
- Destination Popularity

- Market Activity

### Signal Configuration

#### For every source:

- Source
- Geography
- Refresh Frequency
- Weight
- Reliability
- Historical Correlation
- Active/Inactive

### Contextual Impact

#### Example:

#### Dubai Hotel Occupancy:

92%

#### Historical Relationship:

Venue demand +8%

#### Current Estimated Impact:

+6%

### Combined Context

The AI should not treat each factor independently.

#### Example:

**Long Weekend + Hotel Occupancy 94% + Nearby Exhibition + Good Weather could produce:**

Combined Demand Impact: **+23%**

rather than blindly adding percentages.

### Acceptance Condition

TICVAI can incorporate broader tourism, holiday, transport and market conditions into its demand-intelligence layer.

---

## 10.6.7 — AI Demand Forecasting & Booking Curve Studio

### Purpose

Predict future demand at a granular commercial level.

This is the core predictive engine behind intelligent dynamic pricing.

### Forecast Levels

#### Generate forecasts by:

- Venue
- Product
- Event
- Performance
- Date

- Timeslot
- Price Category
- Channel

### Forecast Horizons

#### Support:

- Intraday
- Tomorrow
- 7 Days
- 30 Days
- Event Horizon
- Seasonal Horizon

### Booking Curve

#### Display:

#### Historical Expected Curve

vs.

#### Current Actual Curve

vs.

#### AI Forecast Curve

#### Example

#### Saturday Performance:

#### Expected at T-7:

62% Sold

#### Actual:

74% Sold

#### Forecast Final Occupancy:

**97%**

### Forecast Outputs

#### Provide:

- Demand
- Attendance
- Occupancy
- Sell-Through
- Expected Sell-Out Time
- Revenue
- Conversion
- Remaining Inventory

### Model Inputs

#### Clearly show which signals contributed:

Internal Sales: 35%

Booking Velocity: 20%

Historical Events: 15%

Nearby Exhibition: 10%

Weather: 8%

Market/Tourism: 7%

Competitor: 5%

These should be explanatory contributions rather than necessarily literal model weights.

### **Forecast Confidence**

#### **Display:**

**Confidence: 91%**

#### **with reasons such as:**

- Strong Historical Data
- Stable Booking Pattern
- Reliable External Signals

### **Forecast Error**

#### **Track:**

- MAPE
- Forecast Bias
- Over-Forecast
- Under-Forecast

### **Acceptance Condition**

TICVAI can generate explainable, granular demand forecasts incorporating internal and approved external signals.

---

## **10.6.8 — Price Elasticity & Revenue Response Intelligence**

### **Purpose**

Estimate how customers are likely to respond to different prices.

### **Demand forecasting answers:**

**How much demand do we expect?**

### **Elasticity answers:**

**How will that demand change if we change the price?**

### **Elasticity Analysis**

#### **Analyze:**

- Price
- Demand
- Conversion
- Revenue
- Margin
- Occupancy
- Customer Segment
- Channel
- Time

- Product

## Price Response Curve

### Example:

| Price   | Expected Demand | Conversion | Revenue          |
|---------|-----------------|------------|------------------|
| AED 220 | 10,400          | 8.9%       | AED 2.29M        |
| AED 240 | 10,000          | 8.4%       | AED 2.40M        |
| AED 250 | 9,800           | 8.1%       | AED 2.45M        |
| AED 265 | 9,500           | 7.8%       | <b>AED 2.52M</b> |
| AED 280 | 8,700           | 6.9%       | AED 2.44M        |

## Elasticity by Segment

### Example:

#### Tourist:

Low Price Sensitivity

#### Resident:

Medium

#### Family:

High

#### VIP:

Low

## Elasticity by Context

### Analyze differences by:

- Weekday/Weekend
- Peak/Off-Peak
- Event Proximity
- Channel
- Season
- Venue

## Confidence

### Where insufficient data exists:

Elasticity Confidence: **Low**

AI should not present false precision.

## Recommended Revenue Zone

**Instead of always recommending one exact price, TICVAI may identify:**

**Optimal Revenue Range: AED 258-268****Acceptance Condition**

TICVAI can estimate the relationship between price, demand, conversion and revenue with visible confidence and supporting evidence.

---

**10.6.9 — AI Pricing Recommendation & Explainability Center****Purpose**

Convert all intelligence generated by Board 6 into actionable pricing recommendations. This is the central AI recommendation screen.

**Recommendation Card****Example:****Saturday Evening — Dubai Venue****Current Price**

AED 250

**AI Recommended Price**

**AED 270 ↑ 8%**

**Expected Revenue Impact**

+AED 73,400

**Demand Impact**

−1.6%

**Expected Occupancy**

96%

**Confidence**

91%

**Why AI Recommends This****Show major drivers:****+ Booking Velocity**

31% above forecast

**+ Nearby Exhibition**

60K expected visitors / 1.3 km away

**+ Weather**

Extreme heat expected to increase indoor demand

**+ Occupancy**

84% already sold

**+ Competitor Pricing**

Market median AED 278

**− Price Elasticity**

Moderate conversion sensitivity above AED 275

**Counterfactual Explanation****Show:**

If the nearby exhibition were not occurring, recommended price would be AED 260.

If occupancy were below 70%, recommendation would remain AED 250.

This makes AI much more understandable.

### **Confidence Breakdown**

#### **Example:**

Data Quality: 95%

Forecast Confidence: 92%

Elasticity Confidence: 84%

External Signal Confidence: 91%

#### **Overall:**

**91%**

### **Actions**

#### **Authorized users can:**

- Accept
- Reject
- Modify
- Send to Simulation
- Send for Approval
- Ignore
- Add Comment

Execution belongs downstream.

### **Natural Language Explanation**

#### **Revenue Manager can ask:**

Why are you recommending AED 270?

TICVAI should explain it in business language.

### **Acceptance Condition**

Every AI pricing recommendation is accompanied by understandable evidence, expected commercial impact, confidence and human-governance actions.

---

## **10.6.10 — AI Signal Registry, Data Quality & Model Governance**

### **Purpose**

Govern the complete data and intelligence ecosystem behind AI pricing.

This is critical. Without this screen, Development team could connect many external sources without giving TICVAI proper control over them.

### **Signal Registry**

#### **Display:**

- Signal
- Category
- Provider

- Source
- Internal/External
- Market
- Refresh Frequency
- Last Update
- Freshness
- Reliability
- Historical Correlation
- Status

### Example

| Signal          | Status  | Freshness | Reliability |
|-----------------|---------|-----------|-------------|
| TICVAI Sales    | Healthy | Live      | 99.9%       |
| Weather         | Healthy | 10 min    | 97%         |
| Nearby Events   | Healthy | 1 hr      | 91%         |
| Competitor A    | Delayed | 8 hr      | 83%         |
| Hotel Occupancy | Healthy | Daily     | 88%         |

### Data Quality Rules

#### Monitor:

- Missing Data
- Delayed Data
- Outliers
- Duplicate Data
- Invalid Values
- Unexpected Changes
- Source Failure

### Signal Trust

#### Configure:

- Approved
- Experimental
- Advisory Only
- Blocked

### AI Use Permission

Each signal can be permitted for:

- Forecasting
- Recommendations
- Simulation
- Automated Pricing

This is important.

### For example:

Competitor pricing may be allowed for recommendations but prohibited from direct automated execution.

### Fallback Policy

### For each signal:

**Use Historical Value / Ignore / Substitute / Reduce Confidence / Stop AI Recommendation**

### Model Registry

#### Maintain:

- Model Name
- Version
- **Purpose**
- Deployment Date
- Training Window
- Validation Result
- Owner
- Status

### Model Performance

#### Track:

- Forecast Accuracy
- Bias
- Recommendation Accuracy
- Revenue Performance
- Drift

### Model Drift

#### Alert:

Weekend demand model accuracy declined from 91% to 76% during the last 30 days.

### AI Auditability

#### For every recommendation preserve:

**Data Snapshot → Signals → Model Version → Forecast → Recommendation → Confidence → Explanation**

### Acceptance Condition

TICVAI maintains complete governance, quality control, versioning and traceability over the signals and predictive models used by AI pricing.

## Board 6 — Final Screen Register

| #       | Backend Screen                                       | Core Responsibility               |
|---------|------------------------------------------------------|-----------------------------------|
| 10.6.1  | AI Pricing Intelligence Command Center               | AI opportunity overview           |
| 10.6.2  | Internal Demand & Booking Signal Hub                 | TICVAI transactional intelligence |
| 10.6.3  | Weather Intelligence & Demand Impact Configuration   | Weather intelligence              |
| 10.6.4  | Nearby Event, Exhibition & Local Demand Intelligence | Local-event intelligence          |
| 10.6.5  | Competitor Pricing & Market Position Intelligence    | Competitor intelligence           |
| 10.6.6  | Market, Tourism, Holiday & Contextual Signal Hub     | Wider external context            |
| 10.6.7  | AI Demand Forecasting & Booking Curve Studio         | Predictive demand                 |
| 10.6.8  | Price Elasticity & Revenue Response Intelligence     | Price-response modeling           |
| 10.6.9  | AI Pricing Recommendation & Explainability Center    | AI recommendations                |
| 10.6.10 | AI Signal Registry, Data Quality & Model Governance  | AI/data governance                |

### Very Important Architecture Boundary

**I recommend we give Development team this separation explicitly:**

## Board 5 — Decision Rules

“Occupancy is 85%, velocity is +30% and T–5 days. According to our rules, what can happen to the price?”

## Board 6 — AI Intelligence

“Why is demand increasing? What is likely to happen next? What price appears commercially optimal?”

## Board 7 — Optimization & Execution

“Let's simulate the recommendation, test alternatives, approve/automate it, deploy it and measure whether it worked.”

**For your exhibition example, the flow becomes:**

**Exhibition detected 1.2 km away**

→ Board 6 evaluates expected attendance, timing, audience relevance and historical correlation

→ **Predicted Demand Impact +17%**

→ Forecast model recalculates demand

→ Elasticity engine evaluates possible prices

→ AI recommends **AED 250 → AED 270** with 91% confidence

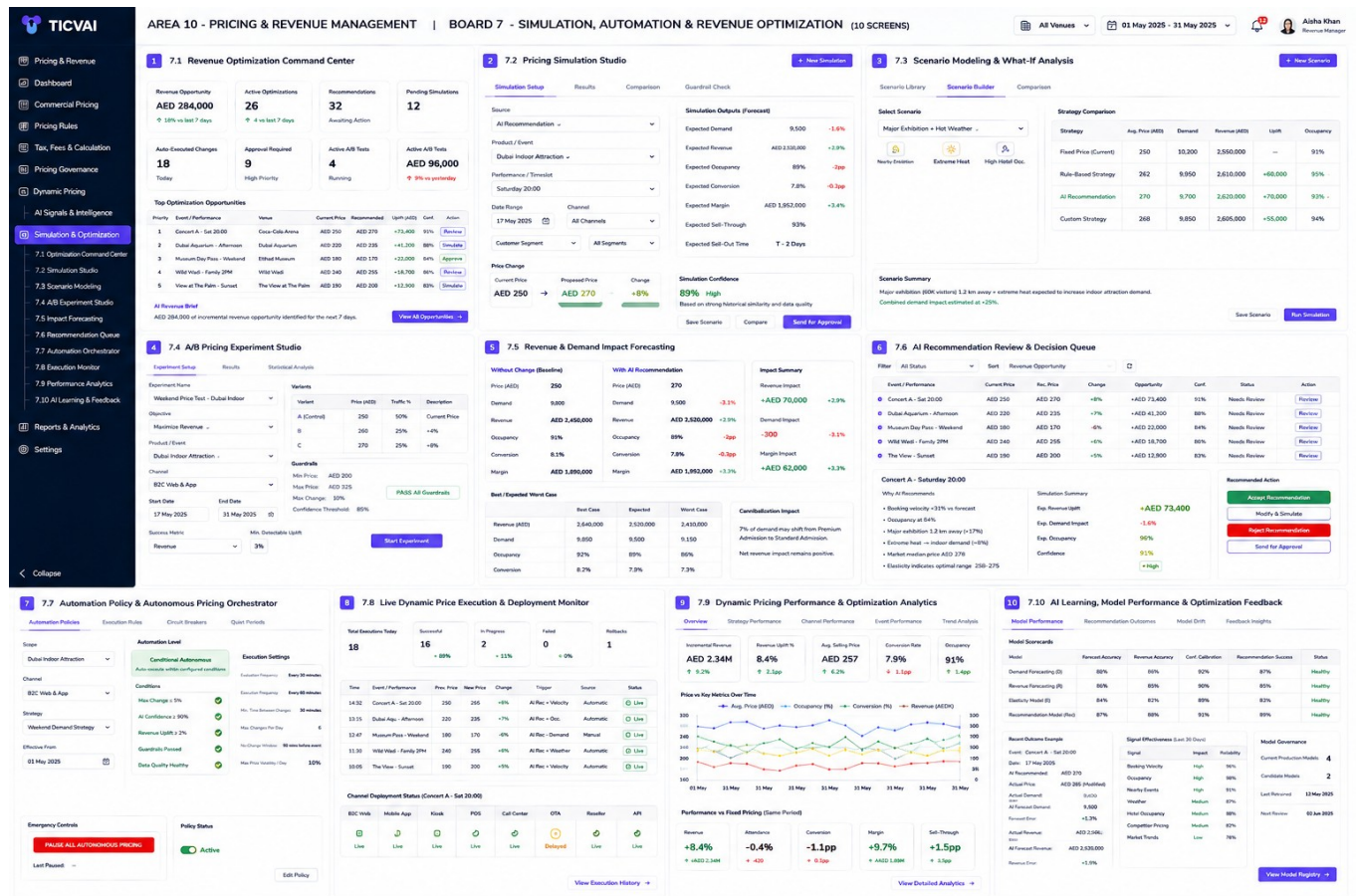
→ Board 5 checks price ladder + commercial guardrails + automation authority

→ Board 7 can simulate and execute the approved strategy.

That is the architecture I would use for TICVAI rather than allowing an AI model to directly modify prices from raw weather/event data.

## Area 10 — Pricing &amp; Revenue Management

## Board 7 — Simulation, Automation &amp; Revenue Optimization



## 10 Backend Configuration Screens — Detailed SOW

Board 7 completes the Dynamic Pricing architecture.

We now have a clean separation:

## Board 5 — Dynamic Pricing Strategy & Rules Engine

### Defines how prices are allowed to move.

## Board 6 — AI Signals, Forecasting & Decision Intelligence

Determines what is happening, what is likely to happen, and what AI recommends.

## Board 7 — Simulation, Automation & Revenue Optimization

Determines whether we should execute the recommendation, how we test it safely, how it goes live, and whether it actually improved the business.

The end-to-end flow becomes:

Signals → Forecast → AI Recommendation → Dynamic Rules → Simulation → Optimization → Approval/Automation → Execution → Monitoring → Performance → Learning

## 10.7.1 — Revenue Optimization Command Center

### Purpose

Provide Revenue Managers with the operational control center for all dynamic pricing simulations, AI recommendations, automated changes, experiments and revenue optimization activity.

### KPI Cards

#### Display:

- Revenue Opportunity
- Incremental Revenue Generated
- Active Optimizations
- Recommendations Awaiting Action
- Pending Simulations
- Auto-Executed Changes
- Approval Required
- Active A/B Tests
- Pricing Exceptions
- Revenue at Risk
- Forecast Accuracy
- Optimization Success Rate

### Optimization Portfolio

#### Each row displays:

- Venue
- Event/Product
- Performance
- Current Price
- Recommended Price
- Forecast Revenue
- Expected Uplift
- Confidence
- Automation Mode
- Approval Status
- Execution Status

#### Example:

| Event        | Current | Recommended | Opportunity | Confidence | Action   |
|--------------|---------|-------------|-------------|------------|----------|
| Concert A    | AED 250 | AED 270     | +AED 73K    | 91%        | Review   |
| Attraction B | AED 220 | AED 235     | +AED 41K    | 88%        | Simulate |
| Event C      | AED     | AED 170     | +AED 22K    | 84%        | Approv   |

| Event | Current | Recommended | Opportunity | Confidence | Action |
|-------|---------|-------------|-------------|------------|--------|
|       | 180     |             |             |            | e      |

### Priority Ranking

#### Prioritize recommendations according to:

- Revenue Opportunity
- Revenue Risk
- Event Proximity
- Confidence
- Inventory Position
- Demand Variance
- Urgency

### AI Revenue Brief

#### Example:

**AED 284,000 of incremental revenue opportunity has been identified for the next seven days. 62% can be executed within existing automation policies.**

#### Quick Actions

- Run Simulation
- Review Recommendations
- Start Experiment
- Approve Changes
- Pause Automation
- Open Execution Monitor

### Acceptance Condition

Revenue teams can monitor and act on pricing optimization opportunities across the organization from one centralized workspace.

## 10.7.2 — Pricing Simulation Studio

### Purpose

Allow any proposed dynamic-pricing change to be tested before affecting live customers. This should become the **sandbox of the Dynamic Pricing Engine**.

### Simulation Sources

#### A simulation can originate from:

- Manual Price Change
- Board 5 Dynamic Rule
- Board 6 AI Recommendation
- New Dynamic Strategy
- Existing Strategy Modification
- Bulk Price Change

### Simulation Scope

**Select:**

- Venue
- Product
- Event
- Performance
- Timeslot
- Price Category
- Channel
- Customer Segment
- Date Range

**Current vs Proposed****Example:****Current Price**

AED 250

**Proposed**

AED 270

**Simulation Outputs****Calculate:**

- Expected Demand
- Expected Conversion
- Expected Attendance
- Expected Occupancy
- Expected Revenue
- Expected Margin
- Expected Sell-Through
- Expected Sell-Out Time
- Average Selling Price

**Comparison****Current Strategy**

vs.

**Proposed Strategy****Confidence****Display:**Simulation Confidence: **89%****based on:**

- Data Volume
- Historical Similarity
- Forecast Confidence
- Elasticity Confidence
- External Signal Quality

**Guardrail Check**

**Before simulation completes, show:**

**Commercial Guardrail — PASS**

**Contract Protection — PASS**

**Price Ladder — PASS**

**Approval Threshold — Revenue Manager**

### **Actions**

- Save Scenario
- Compare Scenario
- Send for Approval
- Start Experiment
- Discard

### **Acceptance Condition**

Authorized users can safely evaluate a pricing recommendation or strategy against forecast commercial outcomes without modifying production prices.

---

## **10.7.3 — Scenario Modeling & What-If Analysis**

### **Purpose**

**Allow Revenue Managers to ask:**

**“What happens if the commercial situation changes?”**

This is broader than Screen 10.7.2, which tests a specific pricing change.

### **Scenario Library**

#### **Support:**

- Low Demand
- Expected Demand
- High Demand
- Sell-Out
- Competitor Price Drop
- Competitor Price Increase
- Extreme Weather
- Rain
- Nearby Exhibition
- Major Concert
- Tourism Surge
- Cancellation Surge
- Inventory Reduction
- Custom Scenario

### **Scenario Inputs**

**Administrators can modify:**

- Demand  $\pm\%$
- Occupancy

- Inventory
- Booking Velocity
- Weather Impact
- Nearby Event Impact
- Competitor Price
- Conversion
- Price
- Event Proximity

### Example

#### Scenario: Major Exhibition + Hot Weather

##### Nearby Exhibition:

+17% Demand

##### Extreme Heat:

+8% Indoor Demand

##### Booking Velocity:

+22%

##### Remaining Capacity:

24%

### Compare Multiple Strategies

#### For the same scenario compare:

##### Fixed Price

vs.

##### Board 5 Rule Strategy

vs.

##### AI Recommendation

vs.

##### Custom Strategy

### Results

| Metric     | Fixed  | Rule-Based | AI           |
|------------|--------|------------|--------------|
| Avg. Price | 250    | 262        | 270          |
| Demand     | 10,200 | 9,950      | 9,700        |
| Revenue    | 2.55M  | 2.61M      | <b>2.62M</b> |
| Occupancy  | 97%    | 95%        | 93%          |
| Margin     | 1.91M  | 1.98M      | <b>2.02M</b> |

**Scenario Save****Allow users to:**

- Save
- Duplicate
- Compare
- Share
- Submit for Review

**Acceptance Condition**

Revenue teams can model alternative market and demand conditions and compare different pricing responses before implementation.

---

**10.7.4 — A/B Pricing Experiment Studio****Purpose**

Allow TICVAI to scientifically test different pricing strategies using controlled customer groups.

This is essential because AI should **learn from actual customer behavior**, not only historical assumptions.

**Experiment Setup****Configure:**

- Experiment Name
- Objective
- Product
- Event
- Performance
- Channel
- Customer Segment
- Start Date
- End Date

**Variant Configuration****Example:****Control A**

AED 250

50% Traffic

**Variant B**

AED 265

50% Traffic

**Multi-Variant Testing****Allow:**

A — AED 250

B — AED 260

C — AED 270

**Success Metrics****Select:**

- Revenue
- Conversion
- Average Selling Price
- Margin
- Sell-Through
- Occupancy
- Revenue per Visitor

**Guardrails****Experiments must respect:**

- Minimum Price
- Maximum Price
- Contract Rates
- Membership Rates
- Regulatory Requirements

**Statistical Confidence****Display:**

- Sample Size
- Confidence Level
- Experiment Duration
- Statistical Significance

**Example Result****Variant B:**

Revenue +7.8%

Conversion –1.2%

ASP +6%

Confidence 96%

**Winner****TICVAI can recommend:**

**Variant B is the statistically preferred strategy.**

But promotion to production follows configured governance.

**Acceptance Condition**

Authorized users can execute controlled pricing experiments and determine statistically whether one strategy performs better than another.

**10.7.5 — Revenue & Demand Impact Forecasting****Purpose**

Provide a dedicated commercial impact assessment before a dynamic-pricing action is approved or executed.

**Baseline**

**Show:****Without Change**

Expected Revenue: AED 2.45M

Expected Demand: 9,800

Expected Occupancy: 91%

**Proposed****With AI Recommendation**

Expected Revenue: AED 2.52M

Expected Demand: 9,500

Expected Occupancy: 89%

**Impact****Revenue:**

**+AED 70K / +2.9%**

**Demand:**

**−300 / −3.1%**

**Margin:**

**+AED 62K**

**Multi-Dimensional Impact****Forecast:**

- Revenue
- Demand
- Attendance
- Occupancy
- Conversion
- Margin
- ASP
- Sell-Through
- Sell-Out Probability

**Customer Impact****Estimate where supported:**

- Member Impact
- Resident Impact
- Tourist Impact
- Family Impact
- B2B Impact

**Cannibalization**

Where multiple products exist, estimate whether increasing/decreasing one price may shift customers to another product.

**Example:**

Increasing Premium Admission to AED 300 may shift approximately 7% of demand toward Standard Admission.

**Sensitivity****Display:**

Best Case / Expected / Worst Case.

**Acceptance Condition**

Approvers can clearly understand expected revenue, demand and customer consequences before authorizing a dynamic pricing action.

---

**10.7.6 — AI Recommendation Review & Decision Queue****Purpose**

Provide the human decision workspace for recommendations produced by Board 6.

This should be the **Revenue Manager's inbox**.

**Recommendation Queue****Display:**

- Recommendation
- Venue
- Event
- Current Price
- Recommended Price
- Change %
- Revenue Opportunity
- Demand Impact
- Confidence
- Urgency
- Governance Level

**Recommendation Detail****Show:****Why AI recommends it**

from Board 6.

**Then add:****Simulation Result**

from Board 7.

**Example:****Current Price:**

AED 250

**Recommended:**

AED 270

**AI Confidence:**

91%

**Simulation Revenue Uplift:**

+AED 73,400

**Demand Impact:**

–1.6%

**Guardrails:**

Passed

**Actions****Authorized users can:**

- Accept
- Reject
- Modify
- Simulate Again
- Schedule
- Send for Approval

**Rejection Reason****Capture:**

- Commercial Judgment
- Brand Positioning
- Customer Sensitivity
- Event Strategy
- Incorrect Signal
- Data Concern
- Other

**Modification****Example:****AI:**

AED 270

**Revenue Manager:**

AED 265

The system records the modification.

**Feedback Loop****The outcome should later be available to the AI-learning layer:**

AI Recommended AED 270

Human Selected AED 265

Actual Revenue Result +5.9%

**Acceptance Condition**

Every material AI recommendation can be reviewed, explained, simulated, approved, modified or rejected with complete traceability.

## 10.7.7 — Automation Policy & Autonomous Pricing Orchestrator

**Purpose**

Control when TICVAI is allowed to execute pricing changes automatically.

Board 5 defines the **strategy-level automation permission**.

This screen manages **operational autonomous execution at scale**.

### Automation Policy

#### Configure:

- Tenant
- Venue
- Product
- Event
- Strategy
- Channel
- Market
- Date/Time

### Automation Levels

#### Advisory

AI recommends only.

#### Human-in-the-Loop

AI recommends → human approval.

#### Conditional Autonomous

Automatically execute only when predefined conditions pass.

#### Autonomous

Continuously optimize within strict approved guardrails.

#### Conditional Example

##### Auto-execute only if:

- Adjustment  $\leq 5\%$
- AI Confidence  $\geq 90\%$
- Revenue Uplift  $\geq 2\%$
- Guardrails Passed
- No Protected Rate
- Forecast Data Healthy

### Frequency

#### Configure:

- Evaluation Frequency
- Execution Frequency
- Minimum Time Between Changes
- Maximum Changes per Day

### Quiet Period

#### Example:

No automated price changes within 60 minutes of event start.

### Circuit Breakers

#### Automatically stop automation when:

- Data Quality Falls

- Model Confidence Drops
- Conversion Drops Abnormally
- Price Volatility Exceeds Threshold
- Revenue Falls
- Integration Fails

### Emergency Control

#### Provide:

#### PAUSE ALL AUTONOMOUS PRICING

with strict permission control.

#### Acceptance Condition

TICVAI can operate controlled autonomous pricing while respecting strategy rules, AI confidence, commercial guardrails and emergency safety controls.

---

## 10.7.8 — Live Dynamic Price Execution & Deployment Monitor

### Purpose

Monitor pricing actions as they are executed across TICVAI's selling ecosystem.

### Execution Feed

#### Each execution shows:

- Timestamp
- Product
- Event
- Previous Price
- New Price
- Trigger
- Strategy
- AI Recommendation
- Approval
- Execution Source
- Status

### Example

**14:32**

Dubai Indoor Attraction

AED 250 → AED 265

#### Trigger:

Booking Velocity + Occupancy

#### AI Confidence:

93%

#### Execution:

Automatic

### Channel Deployment

**Monitor propagation to:**

- B2C
- Mobile App
- POS
- Kiosk
- Call Center
- B2B
- Reseller
- OTA
- APIs

**Status****Queued / Processing / Live / Partial / Failed / Rolled Back****Consistency Monitoring****Example:**

B2C → AED 265 ✓

POS → AED 265 ✓

Kiosk → AED 265 ✓

OTA → AED 250 ⚠

**Alert:****Price inconsistency detected.****Failed Deployment****Actions:**

- Retry
- Hold
- Revert
- Escalate
- Freeze Channel

**Price Timeline****Display every live movement:**

AED 250 → AED 260 → AED 265 → AED 280.

**Board 4 Integration**

Formal publication, rollback and audit governance remains controlled through Board 4.

**Acceptance Condition**

Operations teams can monitor the real-time execution and distribution of dynamic prices across all applicable sales channels.

## 10.7.9 — Dynamic Pricing Performance & Optimization Analytics

**Purpose**

Determine whether dynamic pricing actually improves TICVAI's commercial performance.

**Core KPIs****Measure:**

- Incremental Revenue
- Revenue Uplift %
- Margin Uplift
- Average Selling Price
- Conversion
- Attendance
- Occupancy
- Sell-Through
- Revenue per Available Capacity
- Number of Price Changes
- Strategy ROI

**Comparisons****Support:**

- Dynamic vs Fixed Pricing
- AI vs Rules
- AI vs Human
- Strategy vs Strategy
- Venue vs Venue
- Event vs Event
- Channel vs Channel
- Current vs Previous Period

**Example****AI Dynamic Pricing****Revenue:**

+8.4%

**ASP:**

+6.2%

**Conversion:**

−1.1%

**Attendance:**

−0.4%

**Margin:**

+9.7%

**Price Timeline****Overlay:****Price****with:**

- Demand
- Occupancy

- Conversion
- Booking Velocity
- Revenue

This allows Revenue Managers to understand exactly what happened after each price movement.

### **Recommendation Performance**

#### **Measure:**

- Recommendations Generated
- Accepted
- Rejected
- Modified
- Auto-Executed
- Successful
- Negative Outcome

### **Opportunity Lost**

#### **Important AI feature:**

TICVAI estimates AED 41,000 of revenue opportunity was not captured because three recommendations were rejected.

This should be analytical, not presented as certainty.

### **Drilldown**

#### **Analyze by:**

**Market → Venue → Event → Performance → Product → Price Category → Channel**

### **Acceptance Condition**

TICVAI can quantitatively measure the financial and operational effectiveness of dynamic pricing strategies and AI recommendations.

---

## **10.7.10 — AI Learning, Model Performance & Optimization Feedback**

### **Purpose**

Close the intelligence loop by comparing AI predictions and recommendations with actual commercial outcomes.

This is what allows the system to improve over time.

### **Recommendation Outcome**

#### **For every AI recommendation store:**

#### **Situation**

↓

#### **Signals**

↓

#### **Forecast**

↓

#### **Recommended Price**

↓

**Human/Automation Decision**

↓

**Actual Price**

↓

**Actual Demand**

↓

**Actual Revenue**

↓

**Forecast Error****Example****AI Forecast:**

Demand 9,500

**Actual:**

9,720

**Forecast Error:**

+2.3%

**AI Forecast Revenue:**

AED 2.52M

**Actual:**

AED 2.56M

**Model Scorecards****Track:**

- Demand Forecast Accuracy
- Revenue Forecast Accuracy
- Elasticity Accuracy
- Recommendation Success
- Confidence Calibration
- False Positive Rate
- Revenue Uplift

**Performance by Context****Measure model quality by:**

- Venue
- Product
- Event Type
- Season
- Channel
- Market
- Forecast Horizon

**Model Drift****Detect:**

Weekend pricing model accuracy declined from 92% to 78%.

### Signal Effectiveness

Measure whether external signals are actually useful.

#### Example:

| Signal           | Forecast Contribution | Reliability |
|------------------|-----------------------|-------------|
| Booking Velocity | High                  | 96%         |
| Occupancy        | High                  | 98%         |
| Weather          | Medium                | 87%         |
| Nearby Events    | High                  | 91%         |
| Competitor       | Medium                | 82%         |

### Human Override Analysis

#### Analyze:

Revenue Managers modified 24% of AI recommendations.

Then evaluate whether those modifications improved or reduced outcomes.

This should be used for learning—not employee performance scoring unless explicitly configured and governed.

### Retraining / Model Update

#### Support governed model lifecycle:

**Candidate → Validation → Approved → Production → Monitored → Retired**

Actual model governance remains connected to Board 6's Model Registry.

### AI Optimization Recommendations

#### AI can recommend improvements such as:

Booking velocity should receive greater importance for weekend events.

#### or:

Weather has low predictive value for this indoor venue and should be reduced as an input.

Changes should require governance rather than silently modifying production logic.

### Acceptance Condition

TICVAI continuously evaluates forecast and recommendation quality against actual outcomes and provides governed feedback for improving future pricing intelligence.

## Board 7 — Final Screen Register

| #      | Backend Screen                      | Core Responsibility     |
|--------|-------------------------------------|-------------------------|
| 10.7.1 | Revenue Optimization Command Center | Optimization operations |

| #       | Backend Screen                                         | Core Responsibility            |
|---------|--------------------------------------------------------|--------------------------------|
| 10.7.2  | Pricing Simulation Studio                              | Test proposed pricing          |
| 10.7.3  | Scenario Modeling & What-If Analysis                   | Commercial scenario analysis   |
| 10.7.4  | A/B Pricing Experiment Studio                          | Controlled pricing experiments |
| 10.7.5  | Revenue & Demand Impact Forecasting                    | Pre-execution impact           |
| 10.7.6  | AI Recommendation Review & Decision Queue              | Human-in-the-loop decisions    |
| 10.7.7  | Automation Policy & Autonomous Pricing Orchestrator    | Autonomous execution control   |
| 10.7.8  | Live Dynamic Price Execution & Deployment Monitor      | Live price deployment          |
| 10.7.9  | Dynamic Pricing Performance & Optimization Analytics   | Commercial performance         |
| 10.7.10 | AI Learning, Model Performance & Optimization Feedback | Continuous improvement         |

**Area 10 is now a complete 7-board architecture**

**Board 1 — Commercial Pricing Foundation**  
**Defines the commercial price structures.**

**Board 2 — Pricing Rules & Eligibility**  
**Determines which commercial price applies.**

**Board 3 — Tax, Fees & Calculation Engine**  
**Calculates the final payable amount.**

**Board 4 — Pricing Governance, Publication & Operations**  
 Controls approvals, versions, publication, rollback and audit.

## Board 5 — Dynamic Pricing Strategy & Rules Engine

**Defines how dynamic prices are allowed to move.**

### Board 6 — AI Signals, Forecasting & Decision Intelligence

Understands demand, weather, exhibitions, competitors and other signals and predicts what should happen.

### Board 7 — Simulation, Automation & Revenue Optimization

Tests, executes, monitors and learns from dynamic pricing.

This gives us **70 backend screens for Area 10**, but more importantly, it creates a clean separation between the **commercial pricing engine, dynamic rules engine, AI intelligence engine**, and **revenue optimization/execution layer**. That is the structure I would freeze for Development team.